

# Hysteresis Indices and Beta of IBW sites: Curry, Silverado, and Lake Marguerite

2026-03-26

Aims to plot hysteresis per analyte per storm per site

## Metrics

- *Hysteresis index* is calculated as the median (+/- 95% CI) across 2% intervals of discharge.  
+HI = clockwise (source limitation or nearstream source)  
-HI = counterclockwise (transport limitation or distal source)
- *Beta* is the slope of the CQ relationship on the rising limb of the storm  
CIs on beta are bootstrapped 95% CI

## NOTES

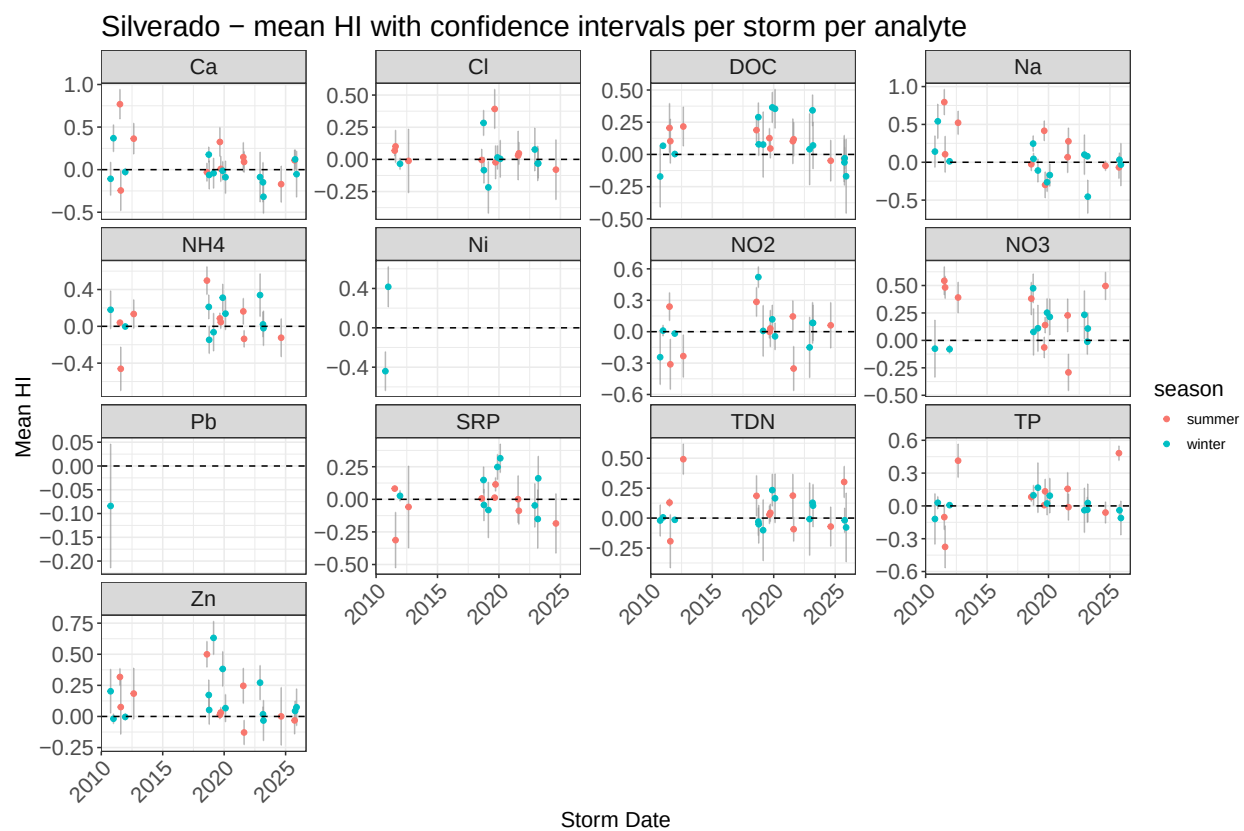
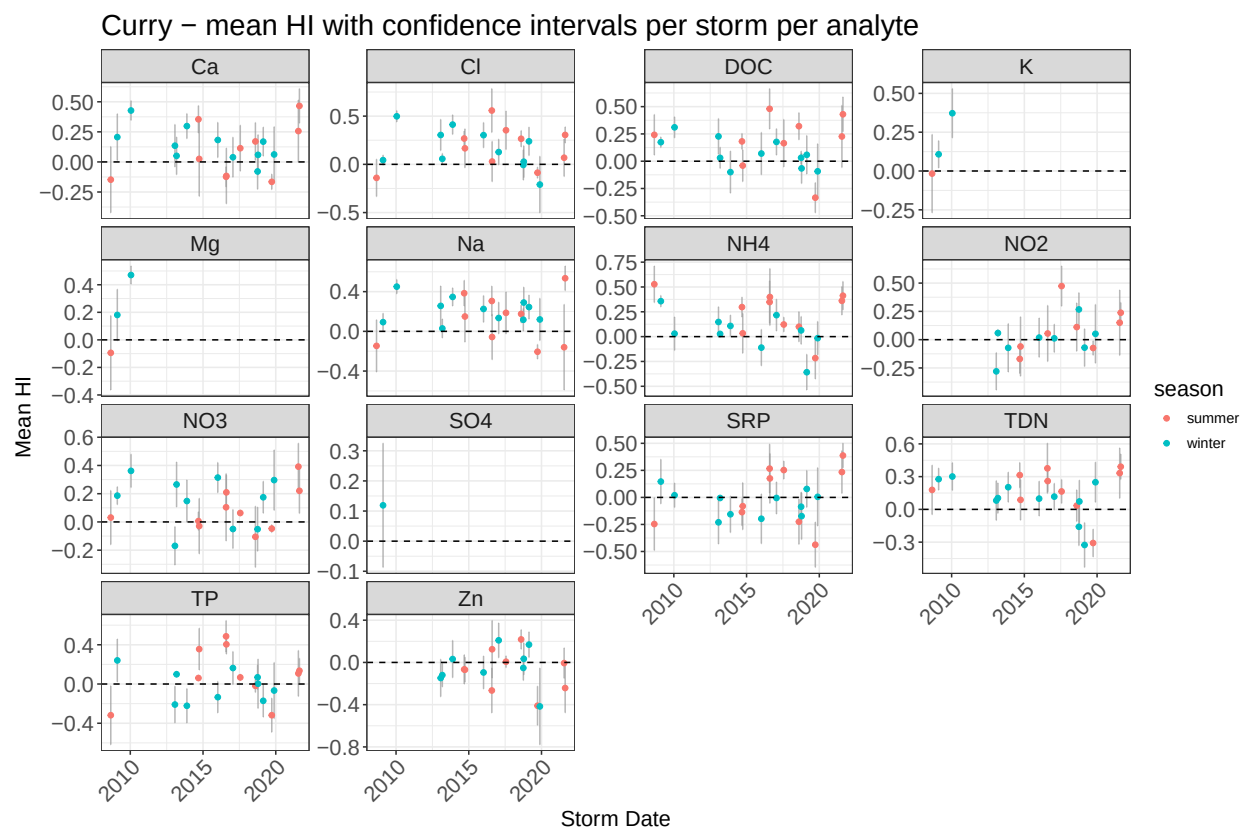
- Lake Marguerite does not have a lot of chems to work with. Do we know if there are more chems than the 624\_runoff\_chems.csv from the dropbox?
- Similarly, there is a gap of data in Silverado 2014-2018, was there less sampling in that time or is the data elsewhere?
- individual analyte outliers > 3sd above mean concentration per storm were removed for hysteresis and beta calculations, but remain for individual storm plotting at the bottom of the report
- Precipitation data was collected from <https://azmet.arizona.edu/azmet/27.htm> - precipitation gage at the Desert Ridge station

## Data munging/cleaning

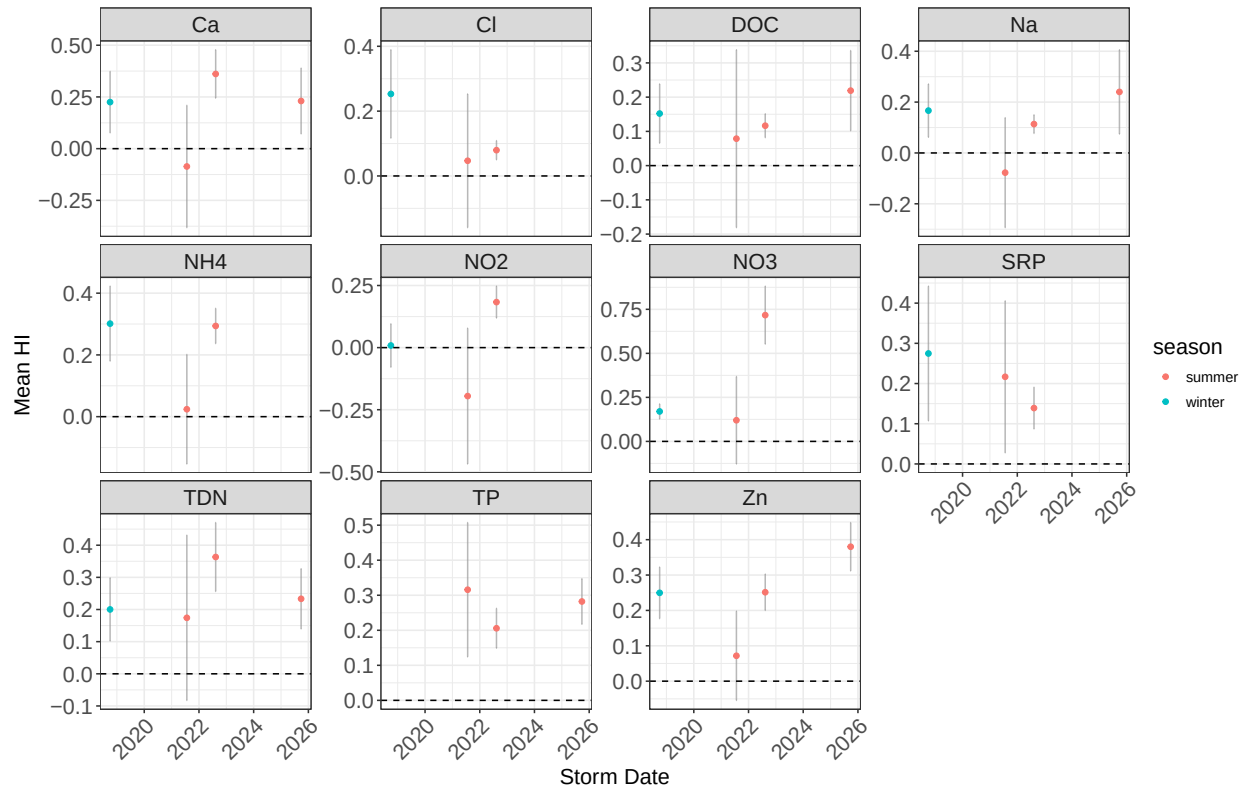
Workflow before this rmd, in order:

- q\_storms.R: import discharge from Lake Margherite, Silverado, & Curry, delineate unique storms, and interpolate Q
- merge\_Q\_chems.R: merge discharge data with chem data per site, produce brief cq plots per analyte per site
- hysteresis\_calc\_2.R: calculate hysteresis based on python function, removed a lot of storms with not enough data. also calculates beta

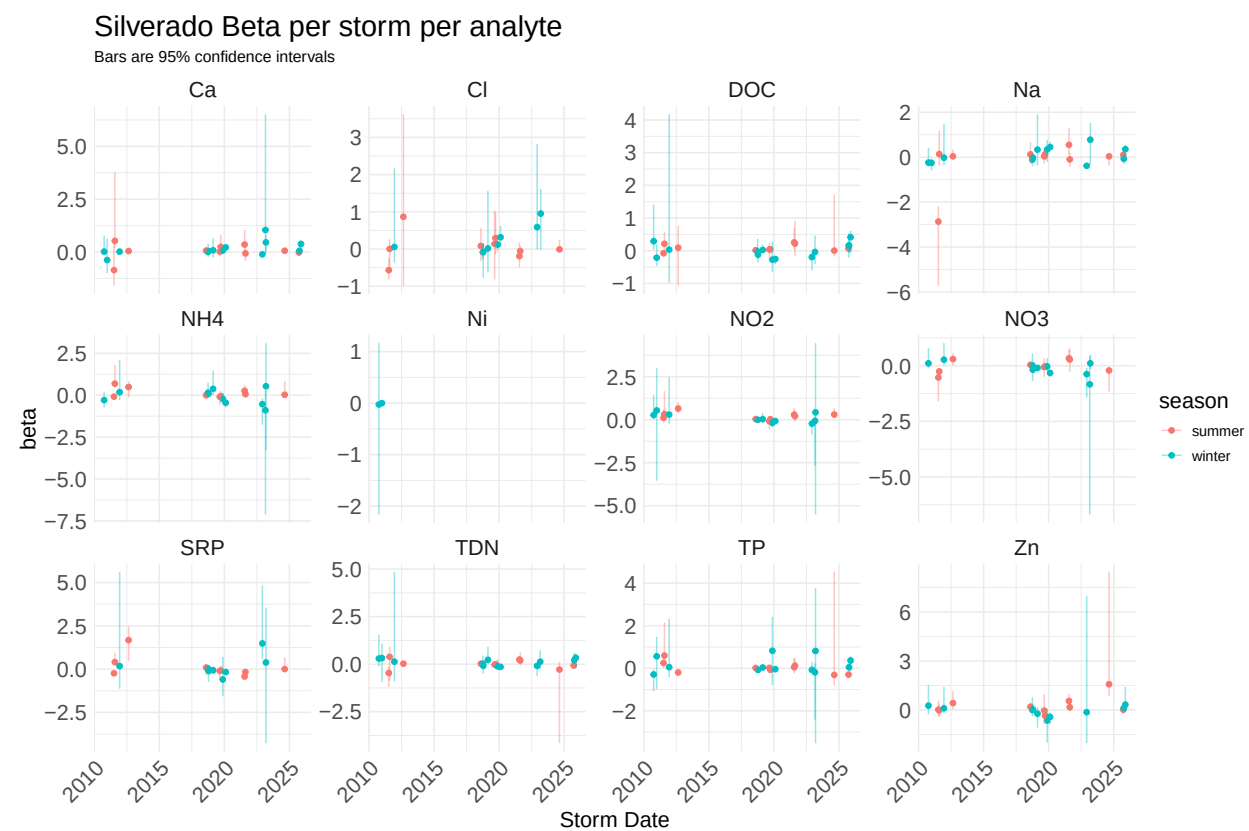
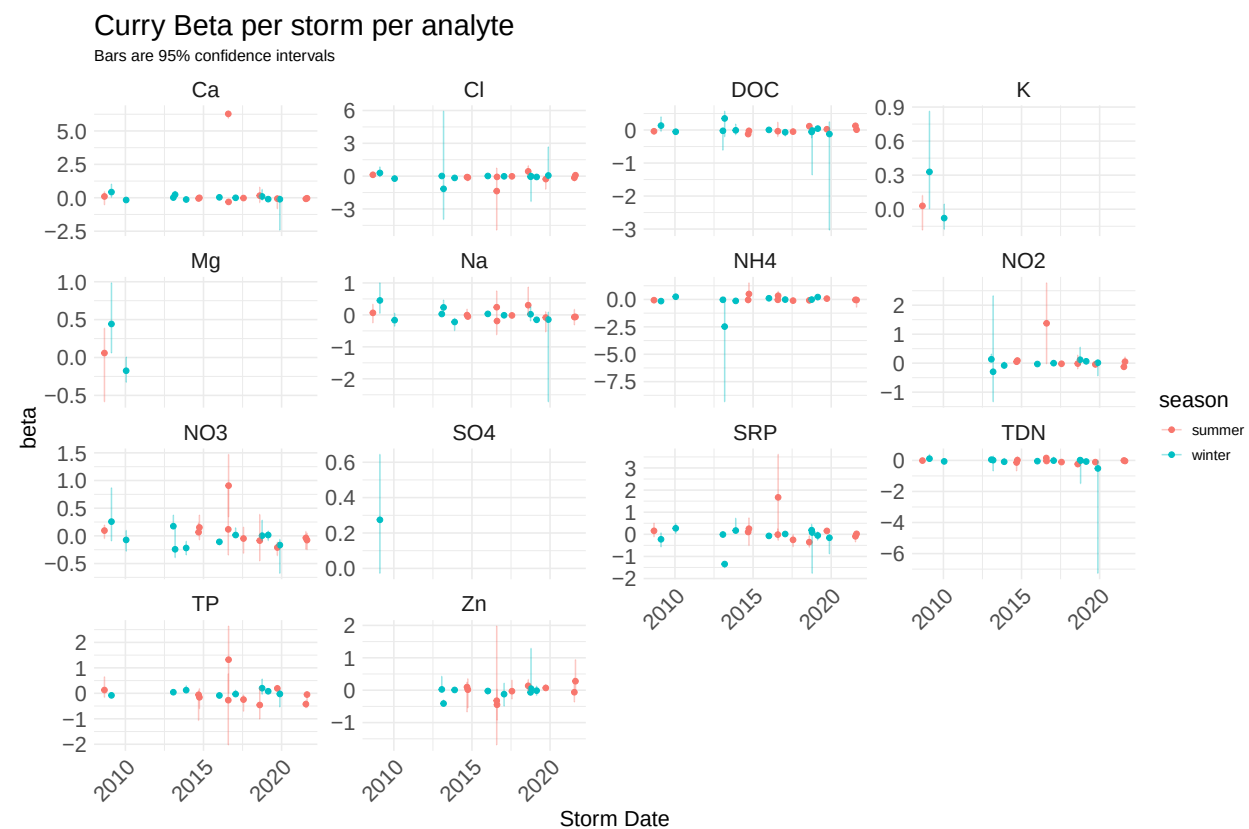
## Plotting Hysteresis



Lake Marguerite – mean HI with confidence intervals per storm per analyte

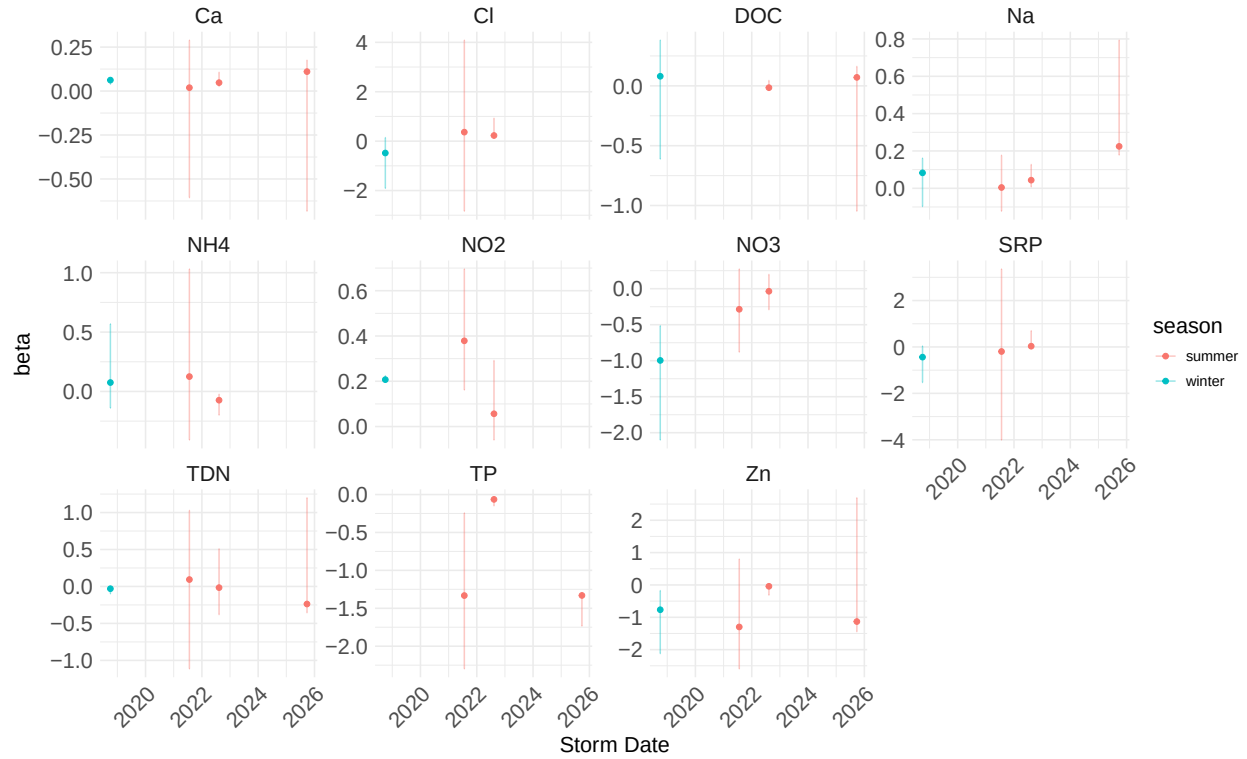


## Plotting Beta



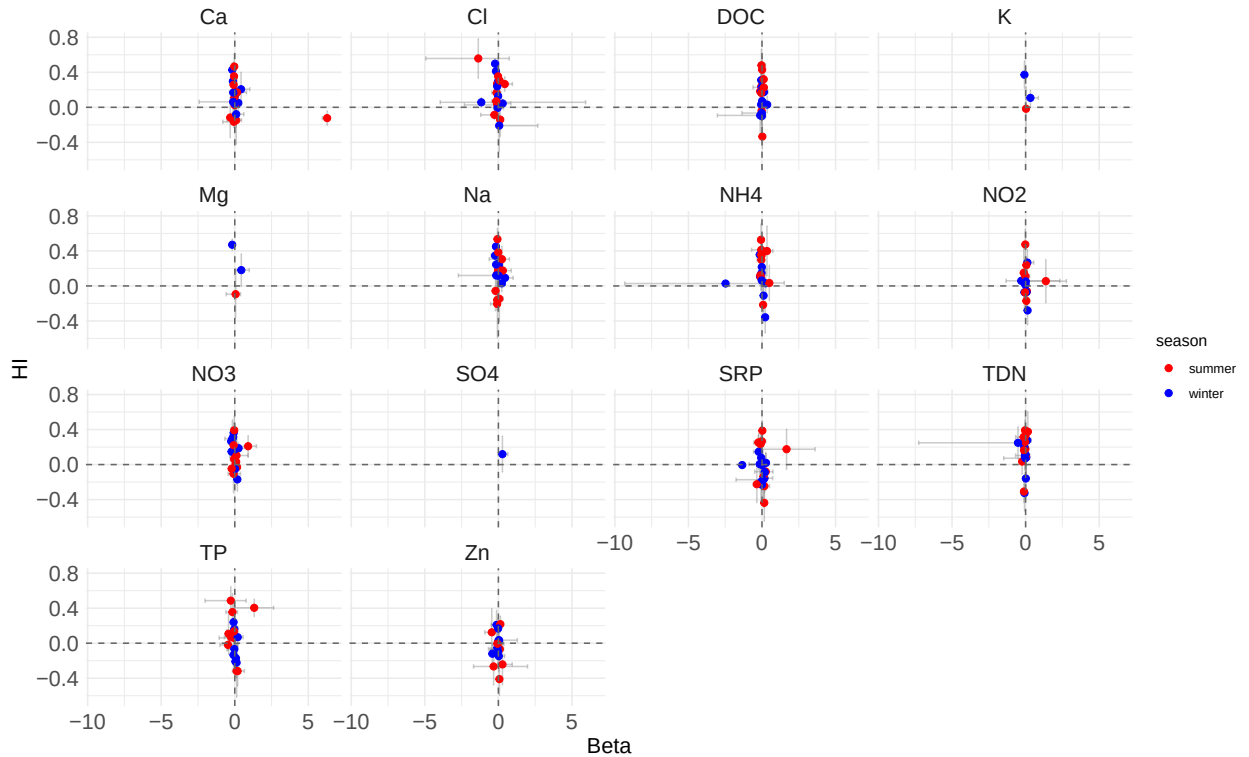
## Lake Marguerite Beta per storm per analyte

Bars are 95% confidence intervals

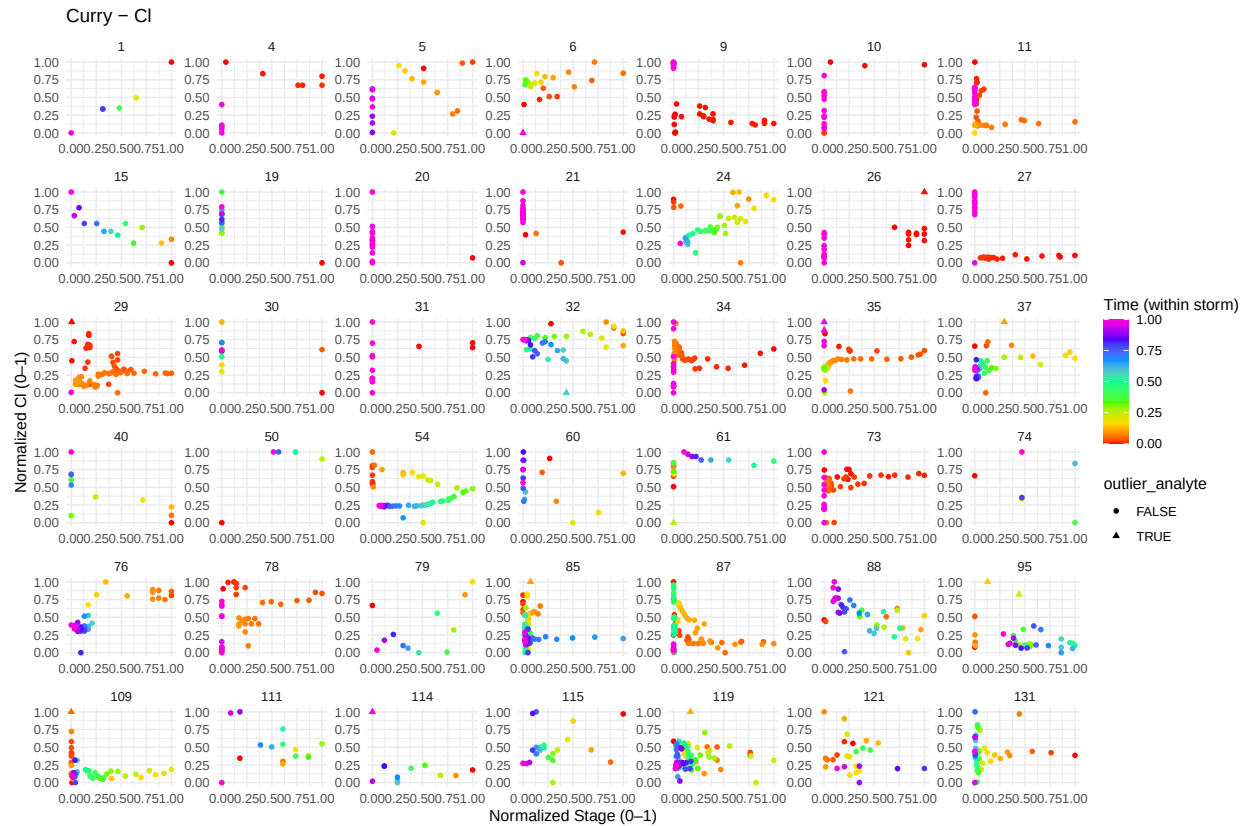
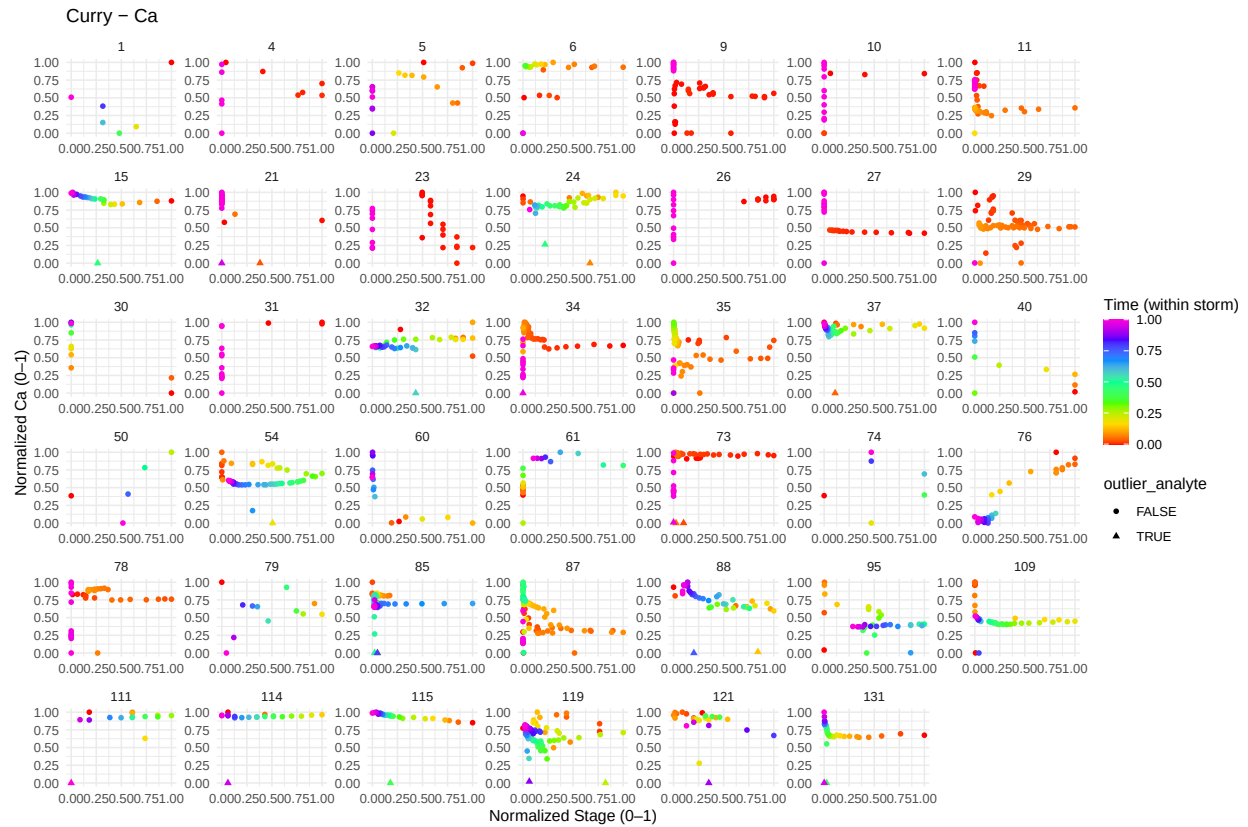


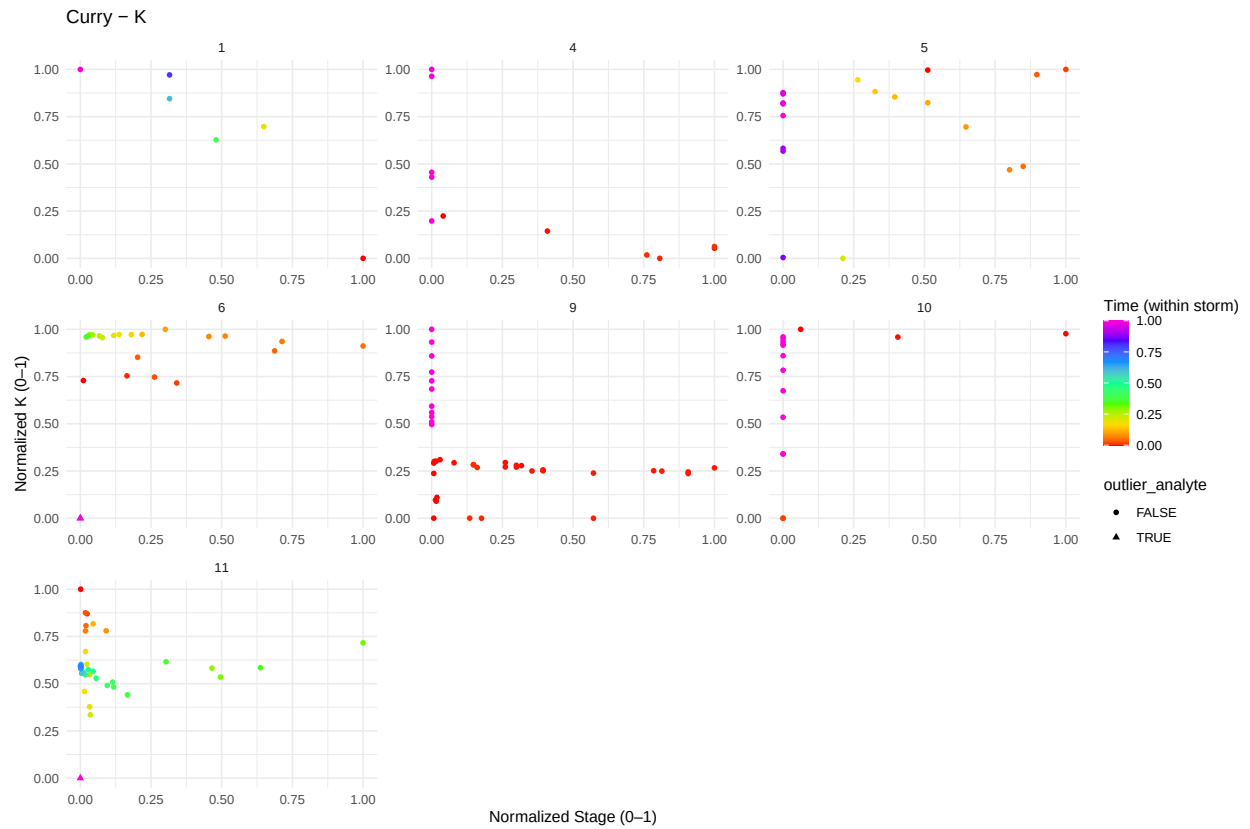
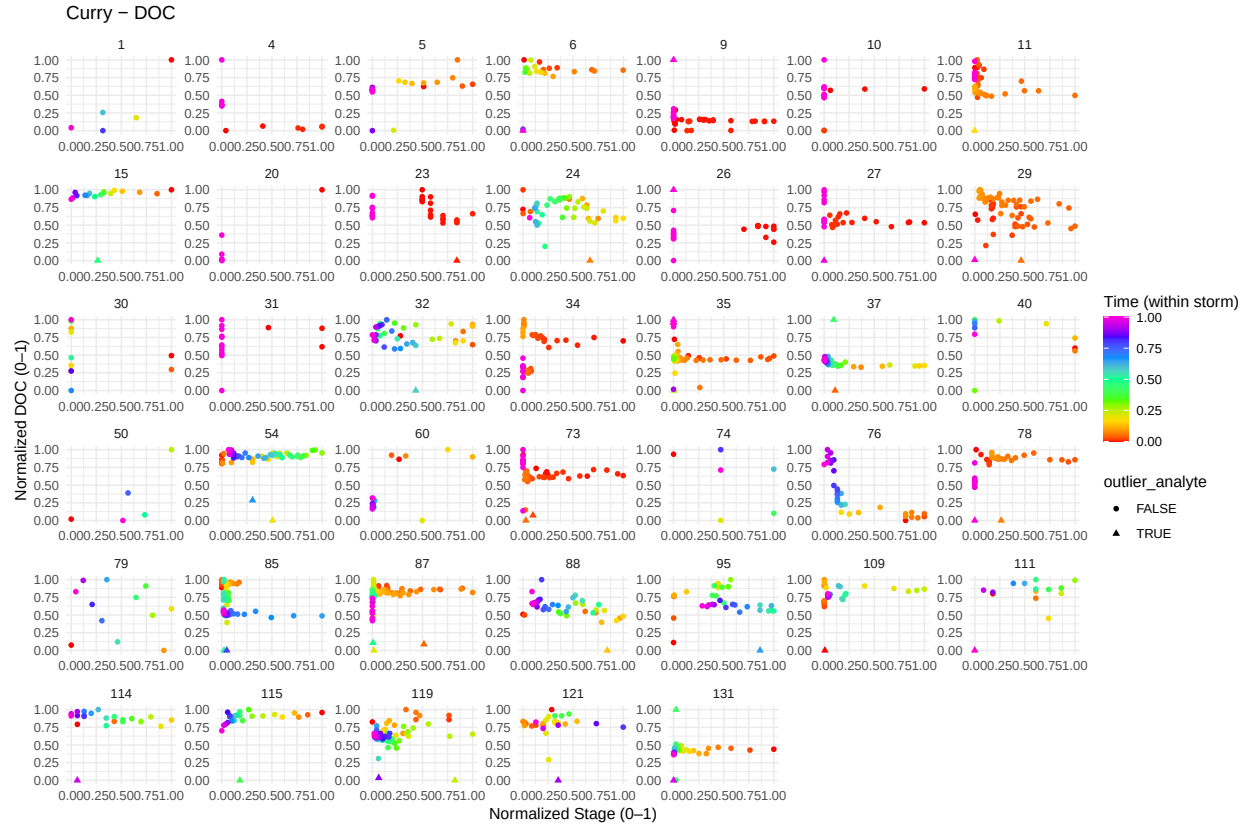
## Curry – HI vs Beta by Analyte

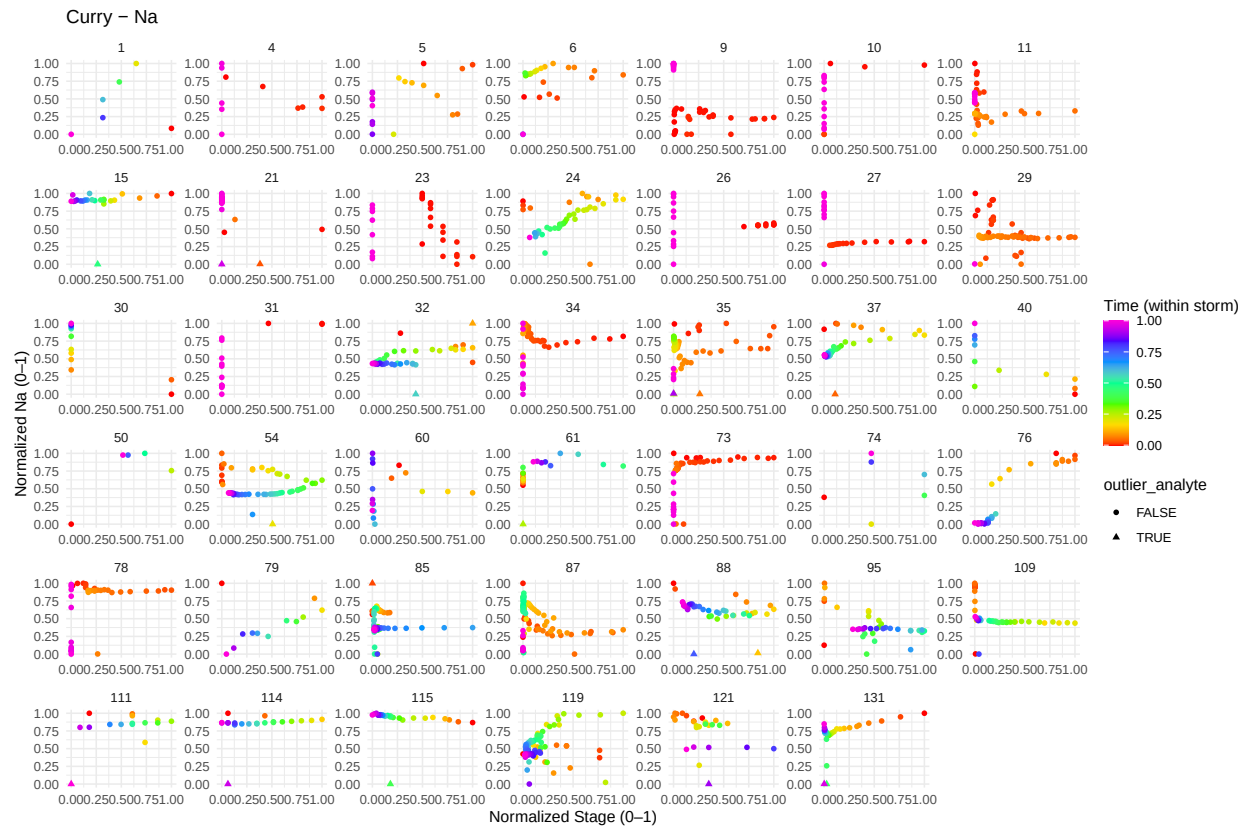
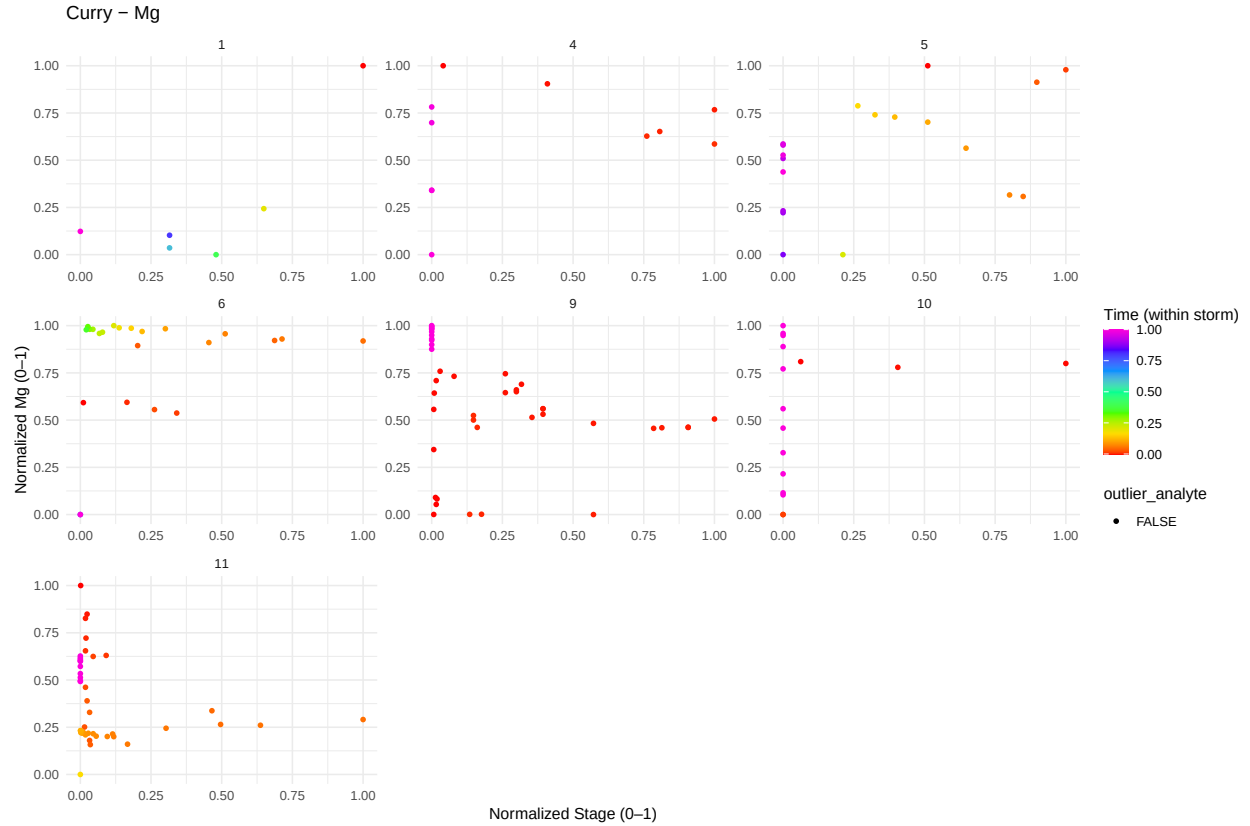
Each dot is an individual storm



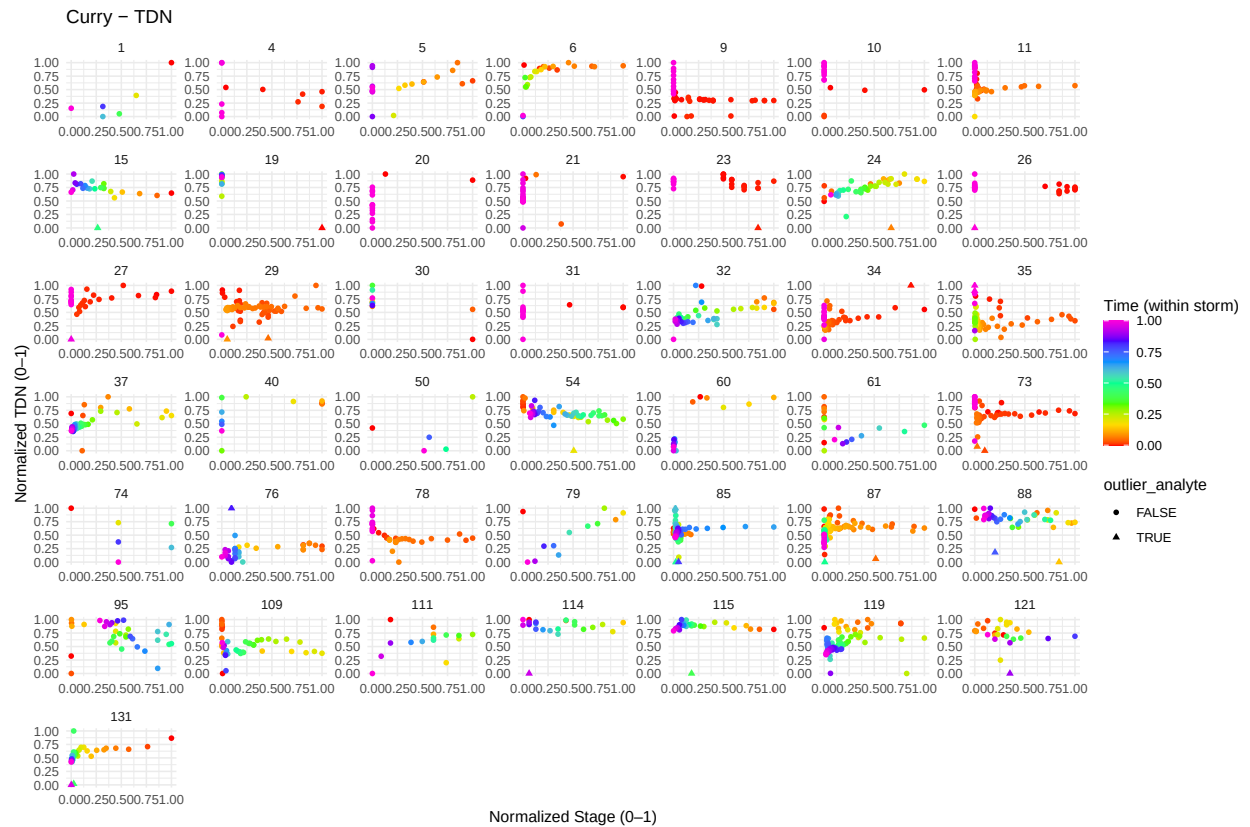
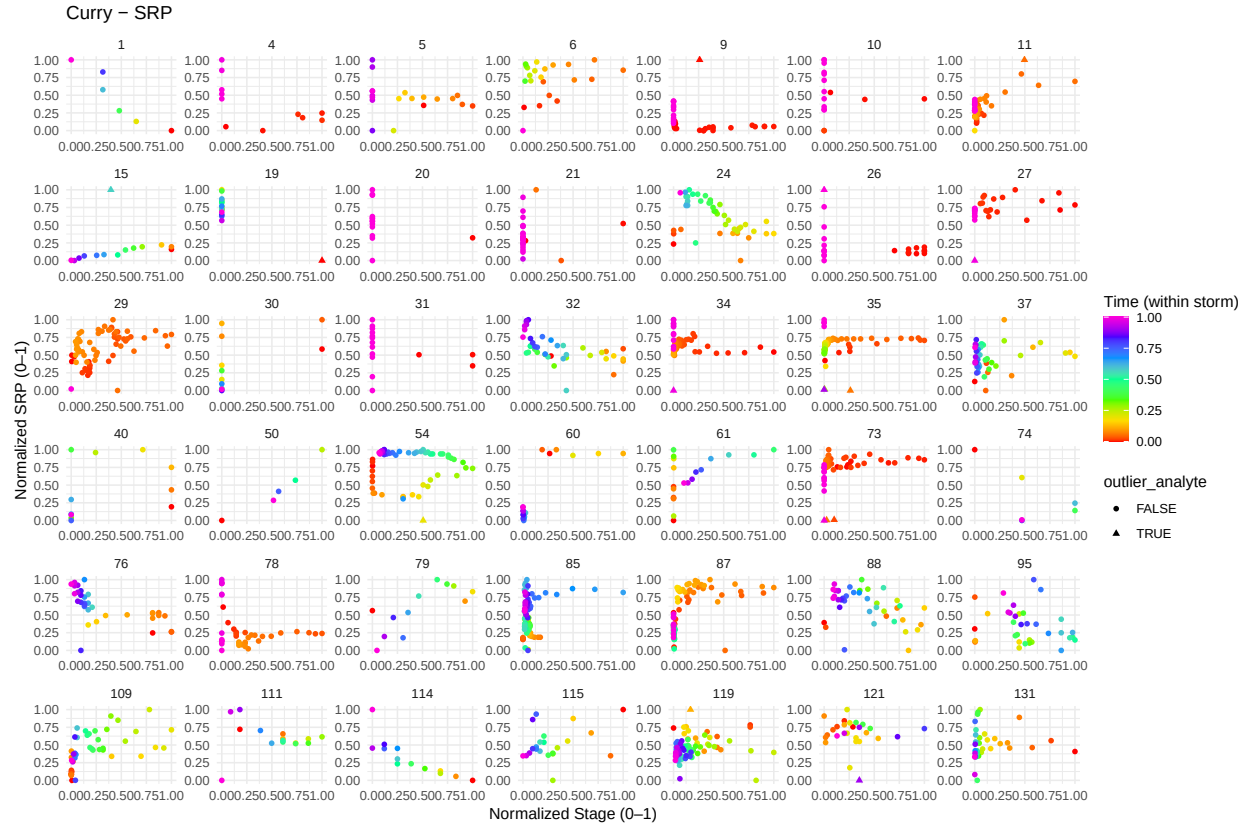
## Plotting CQ

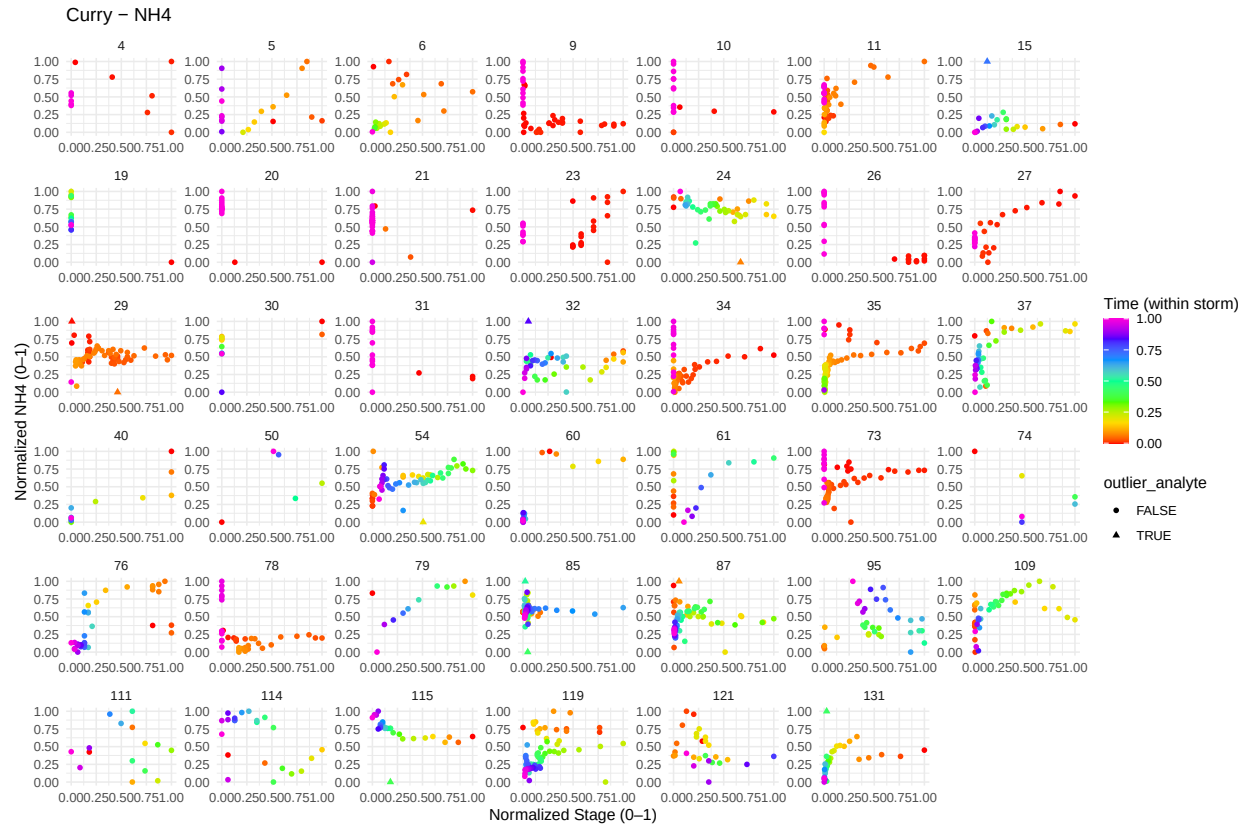
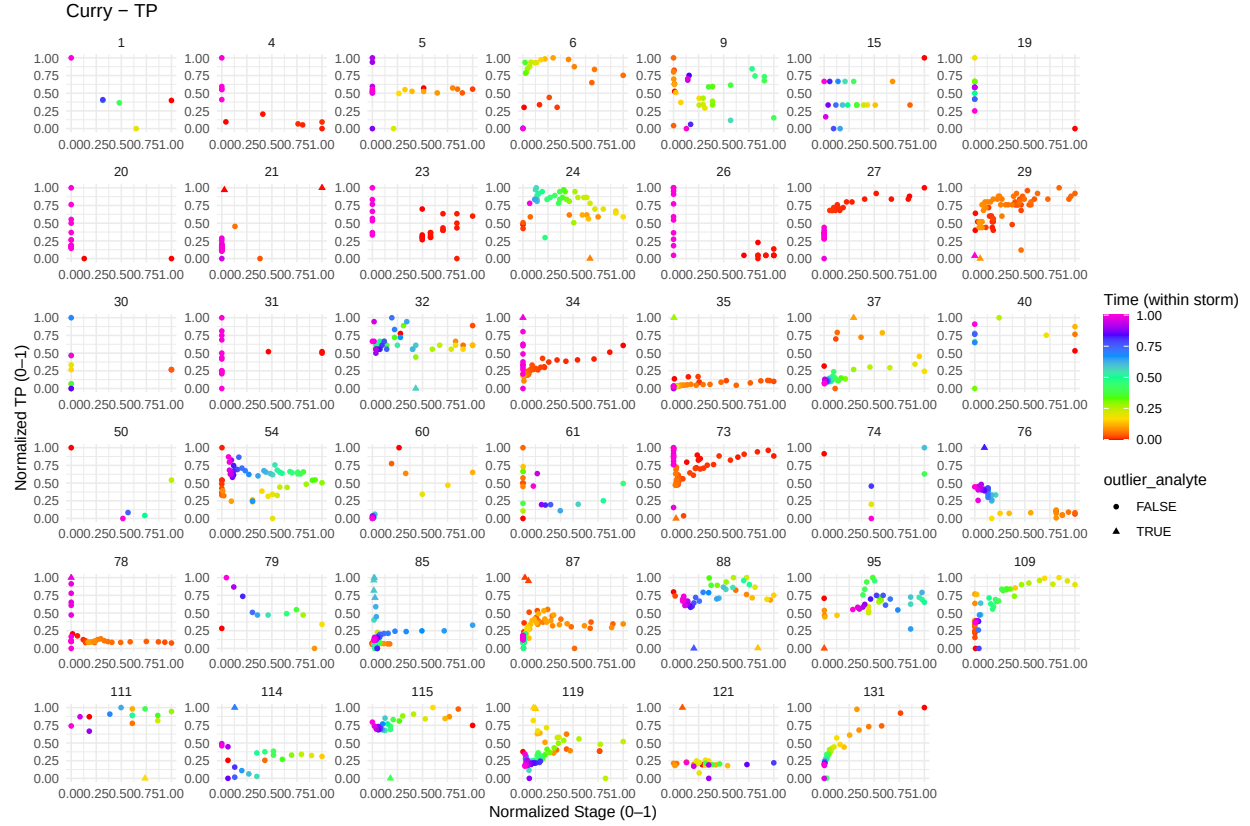


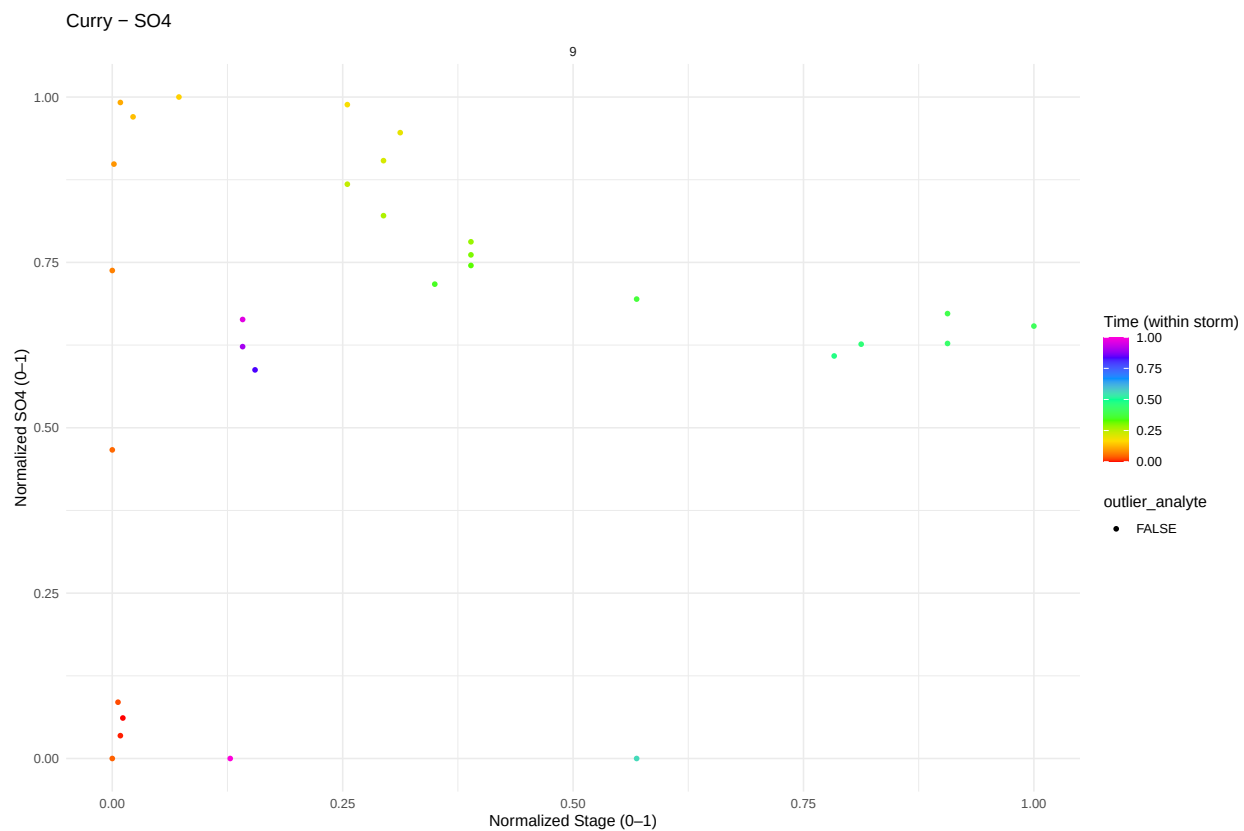
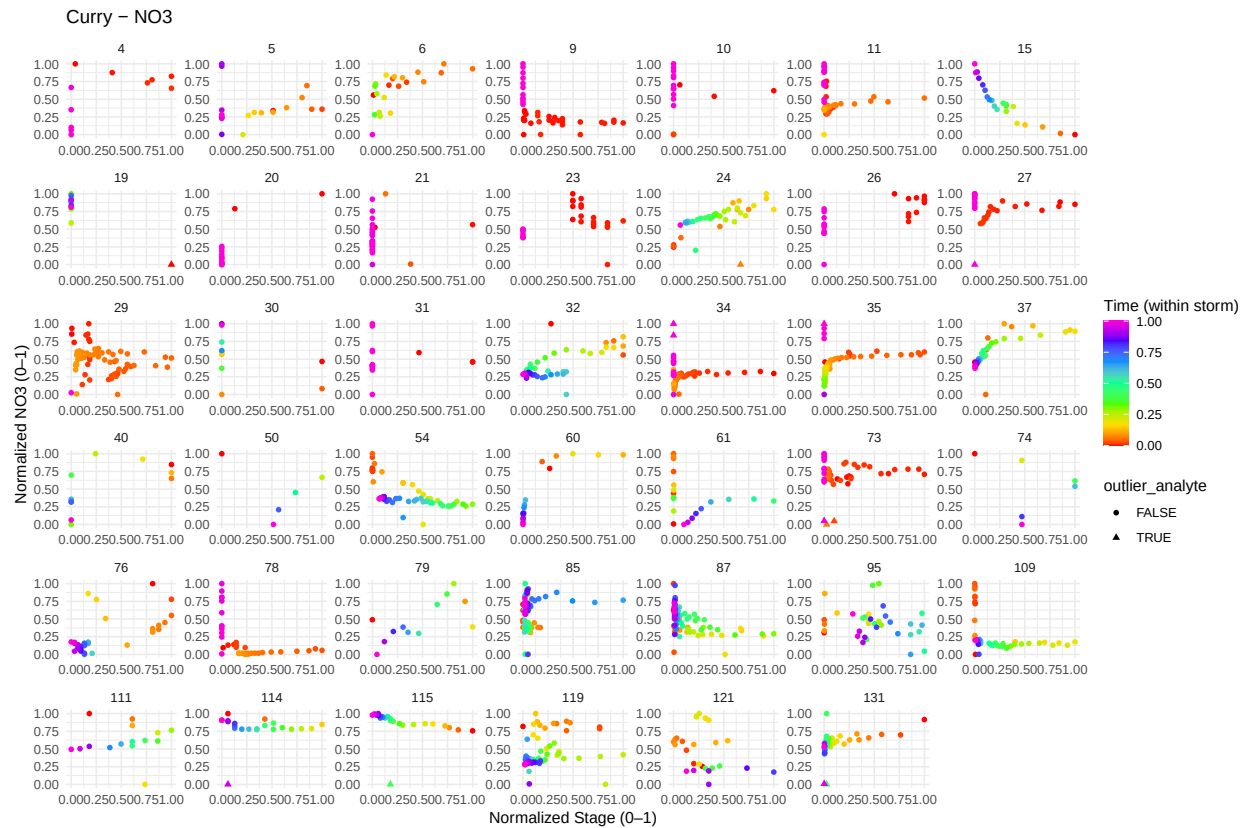


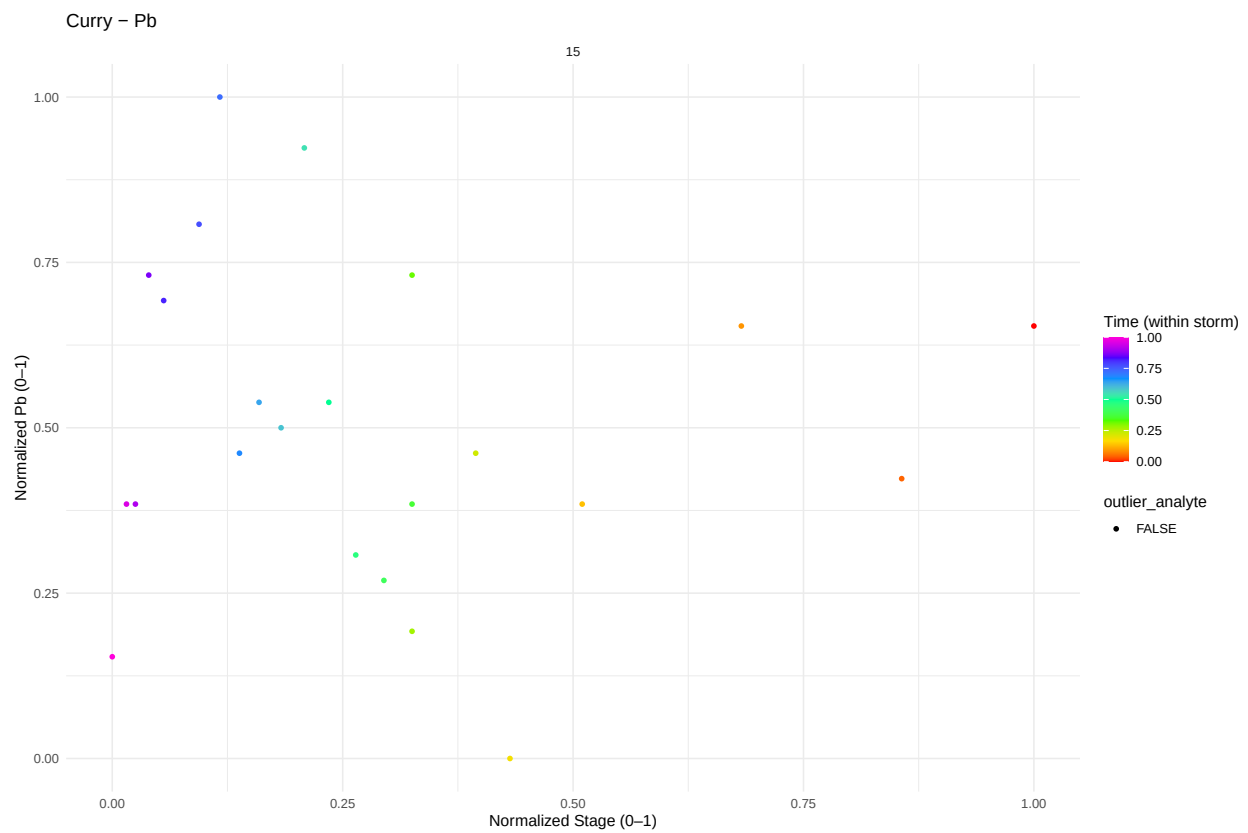
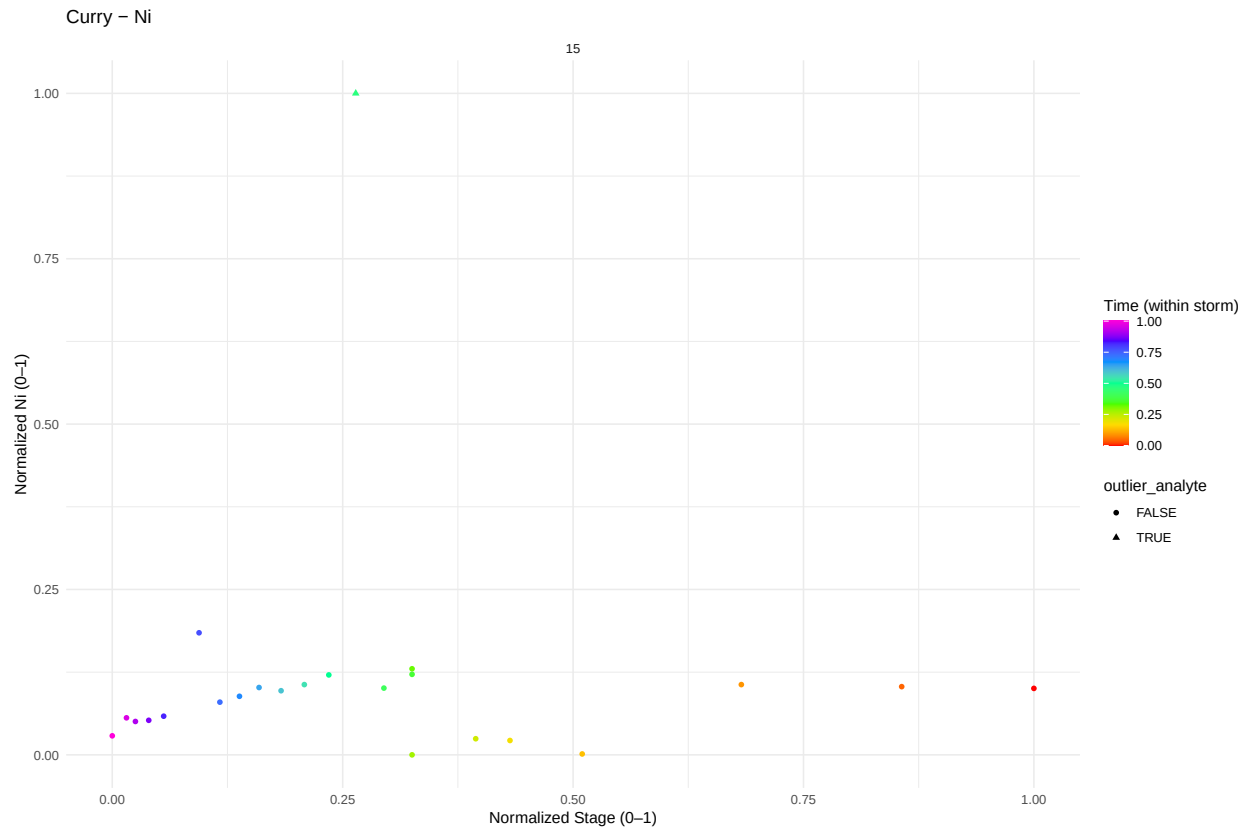


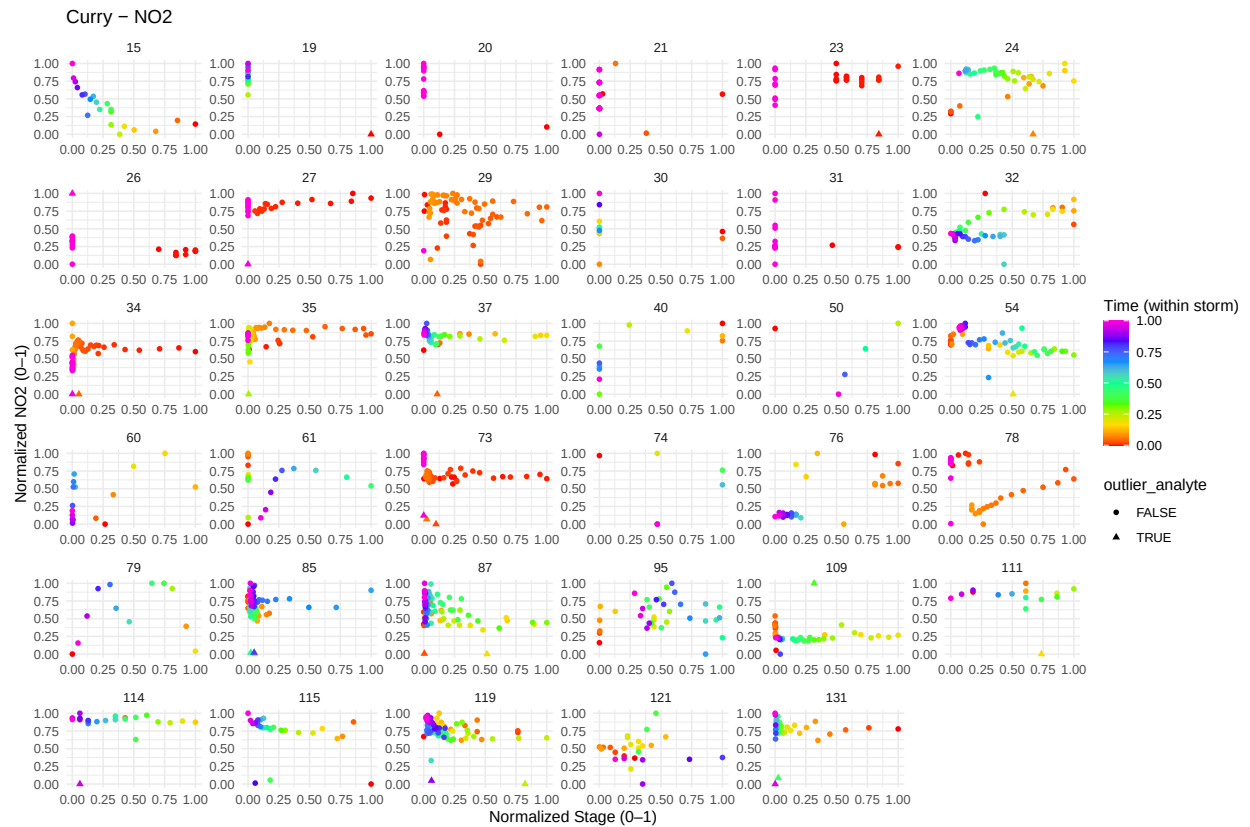
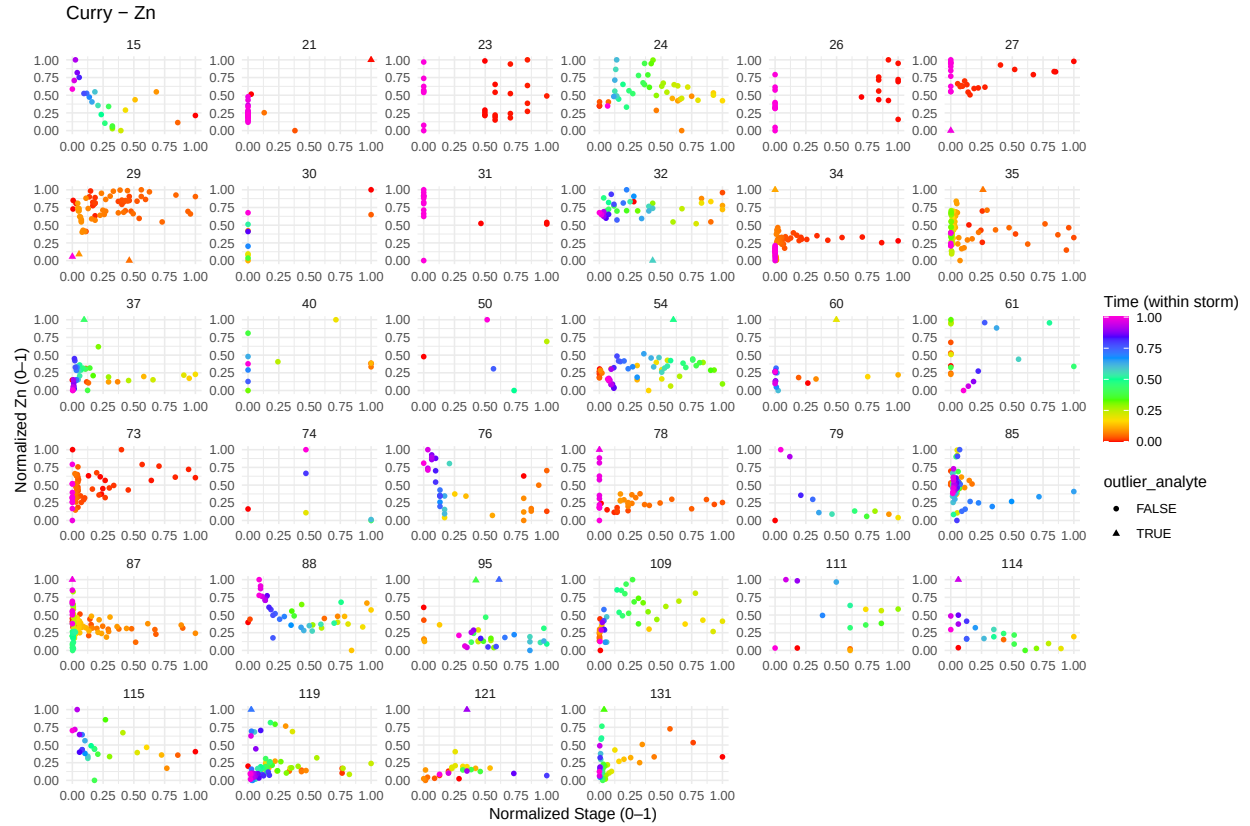


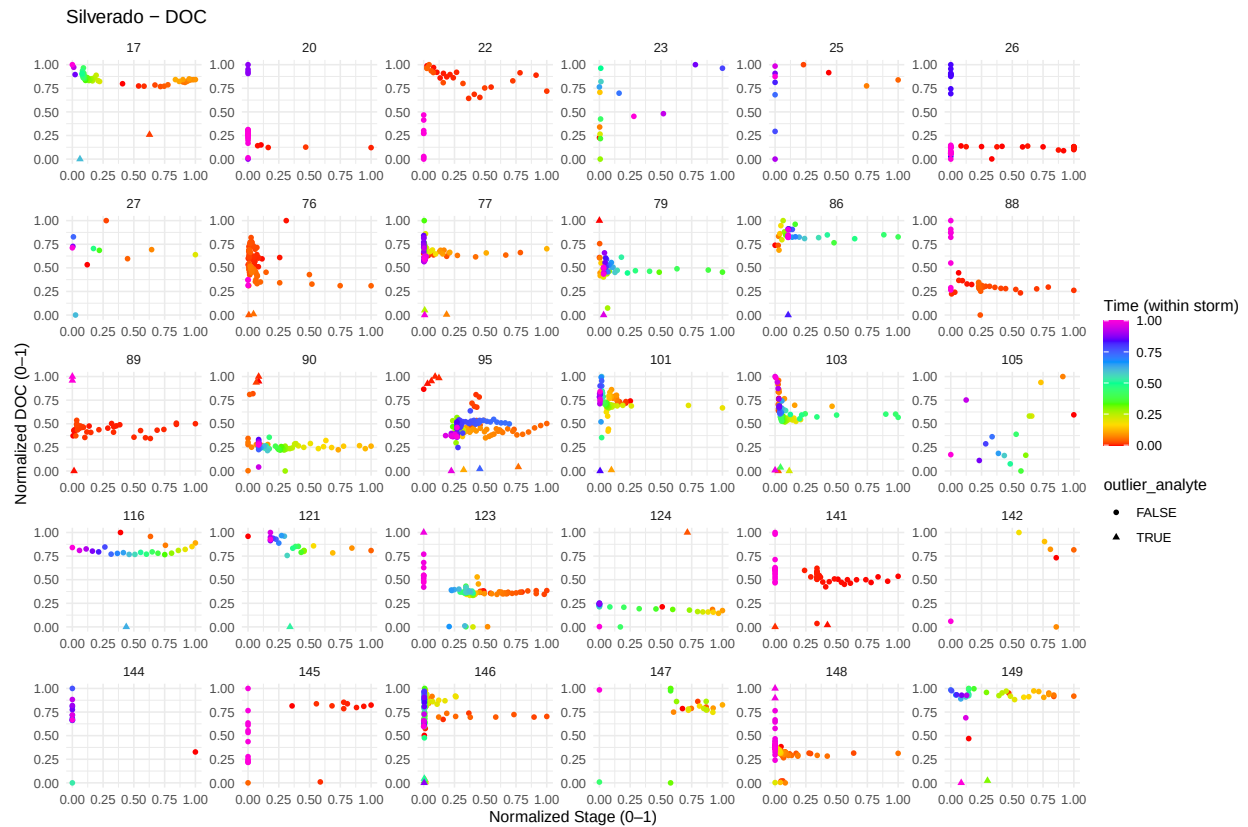
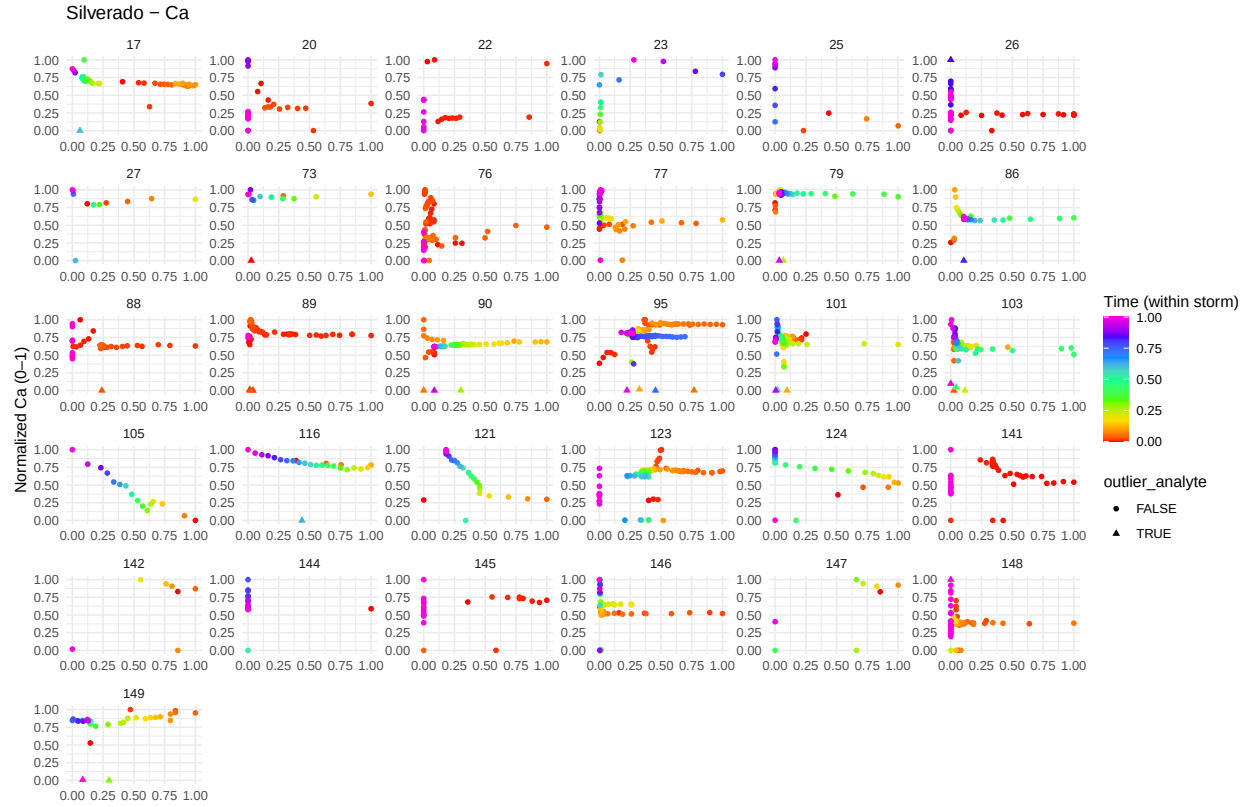


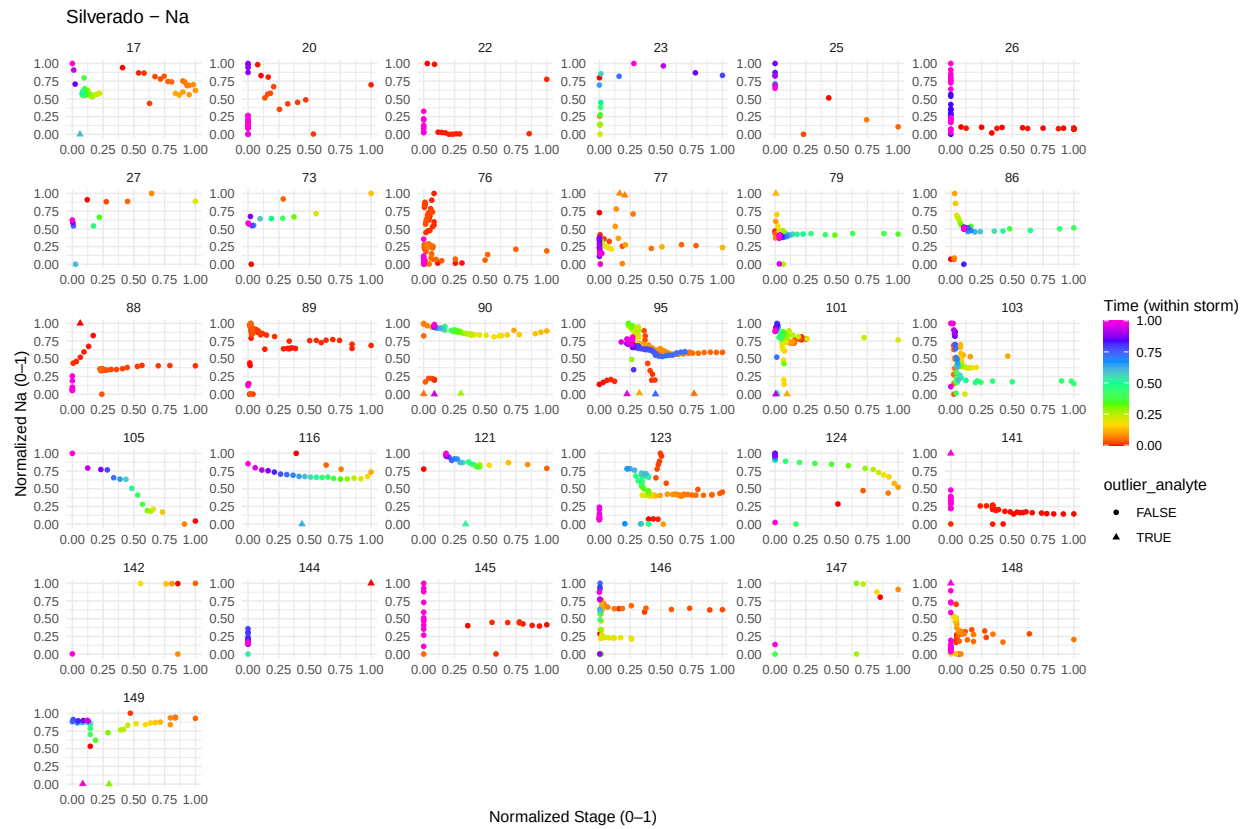
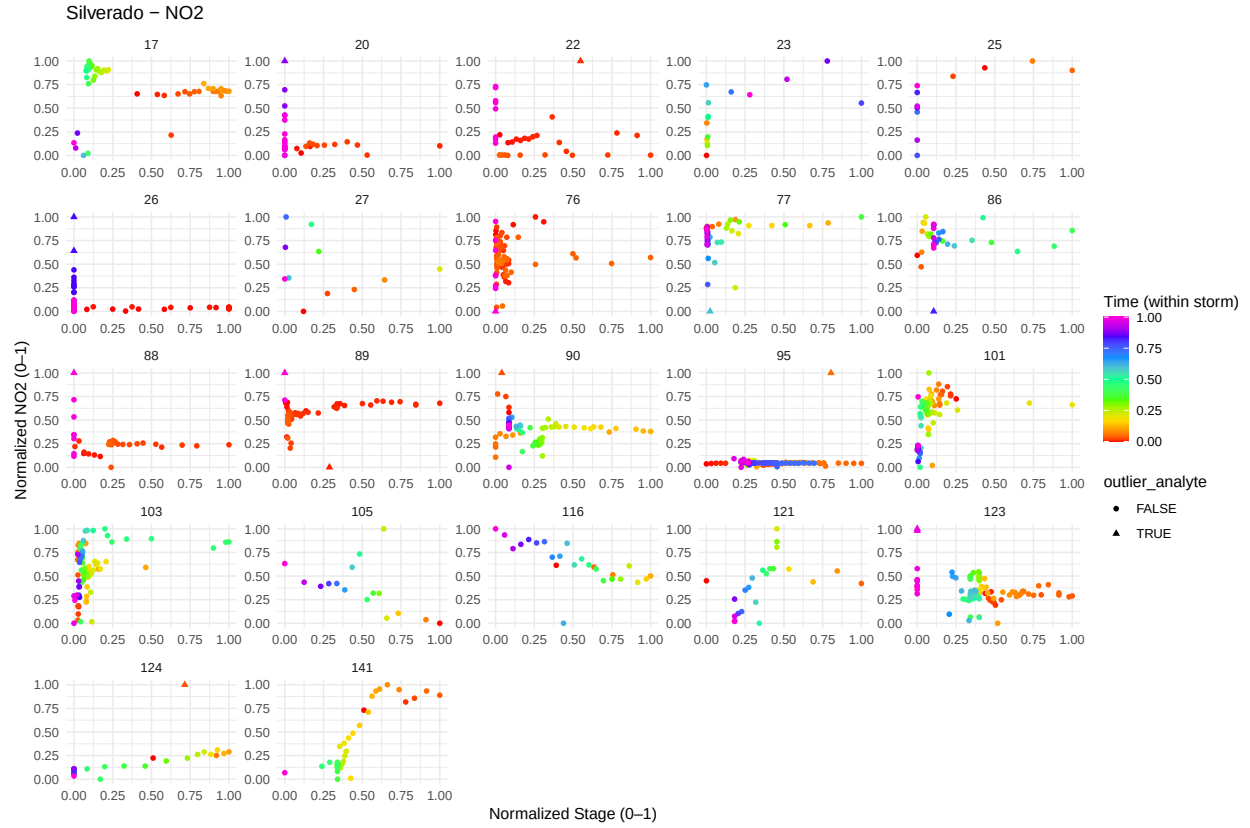


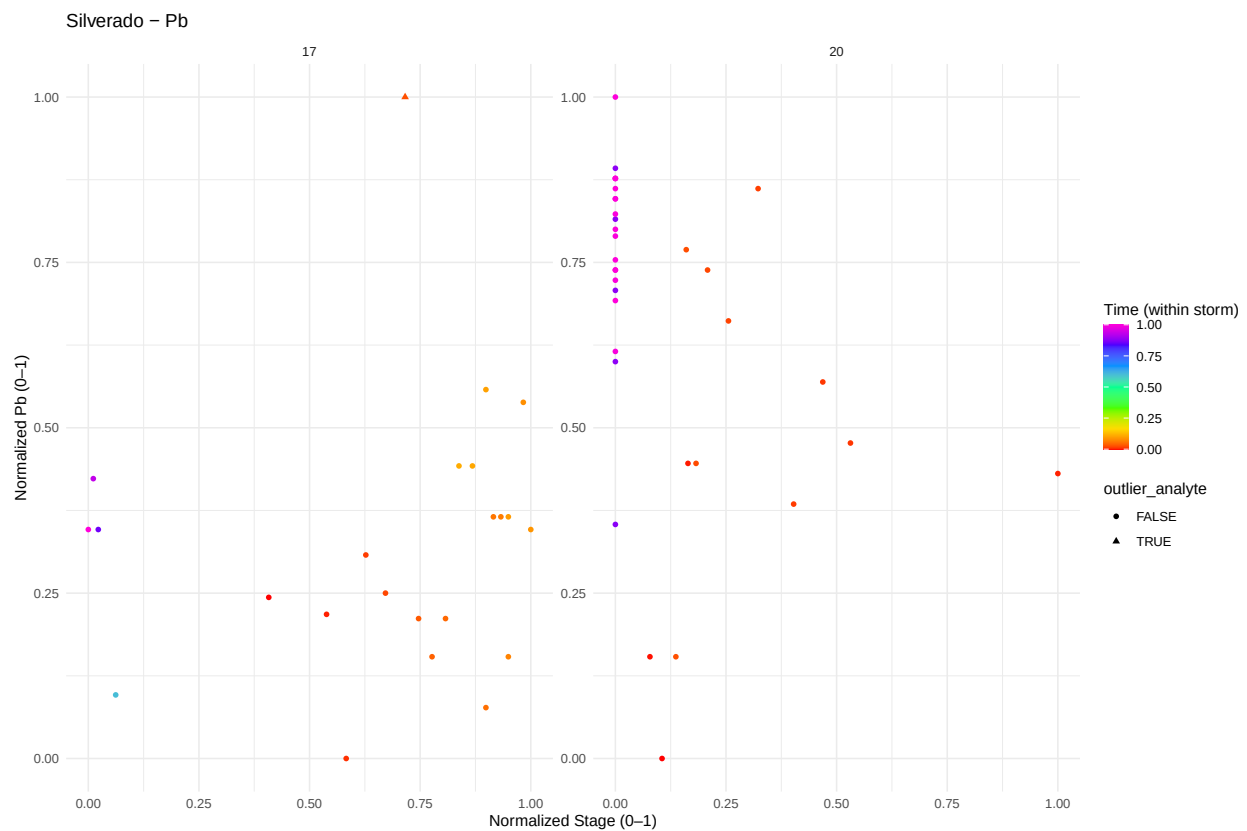
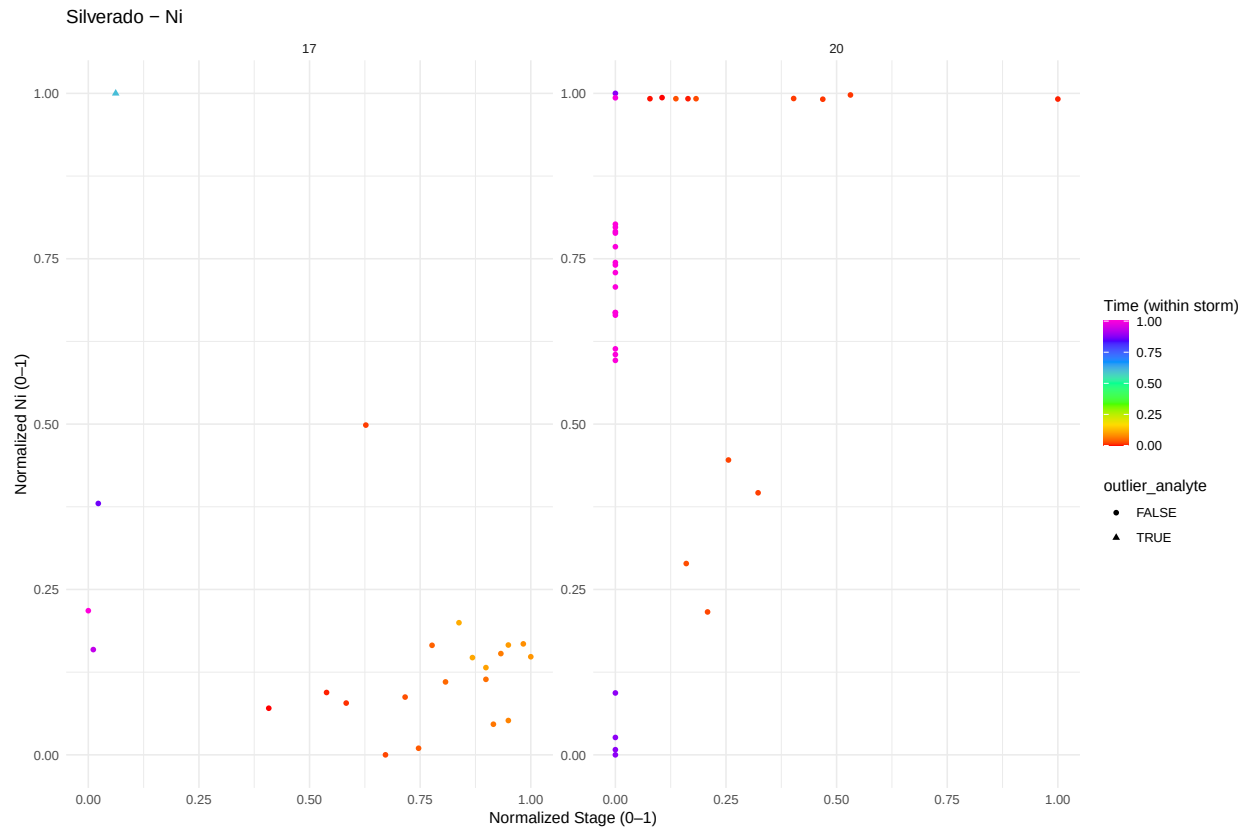




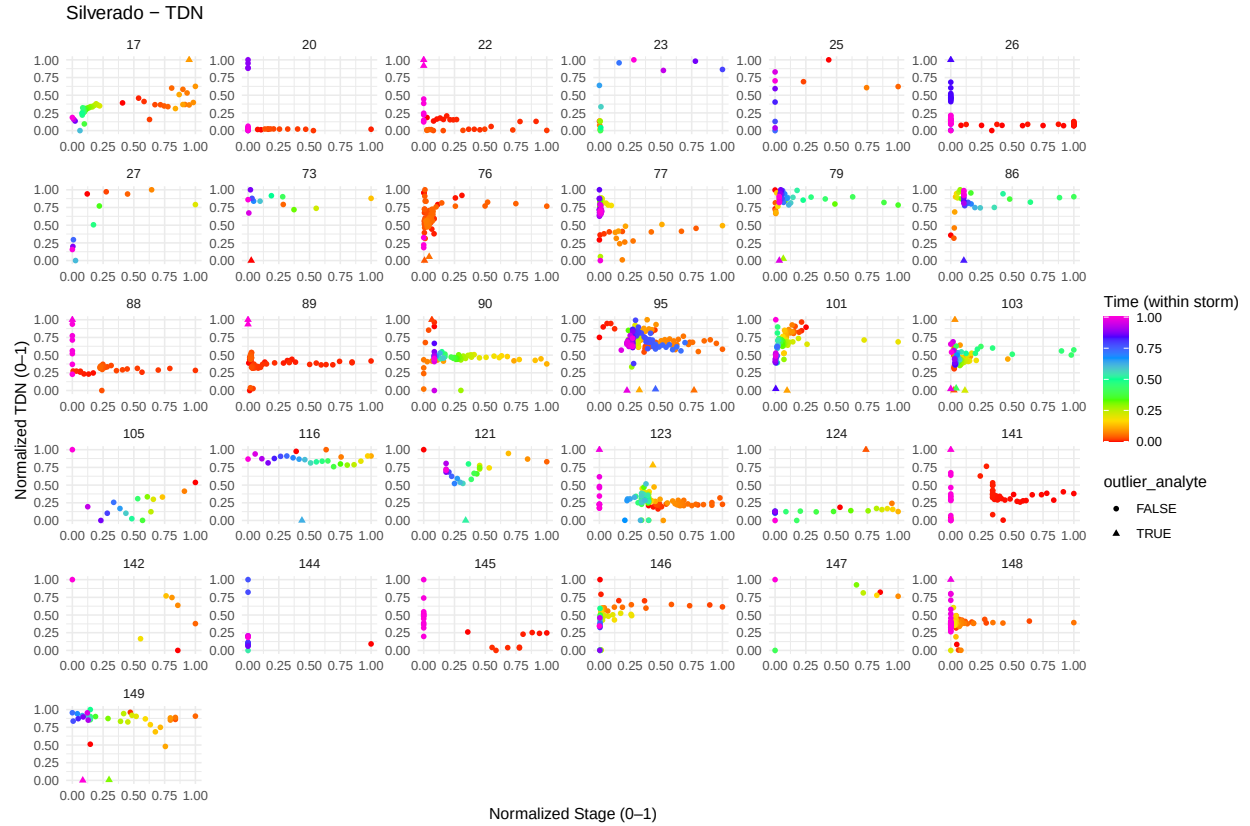


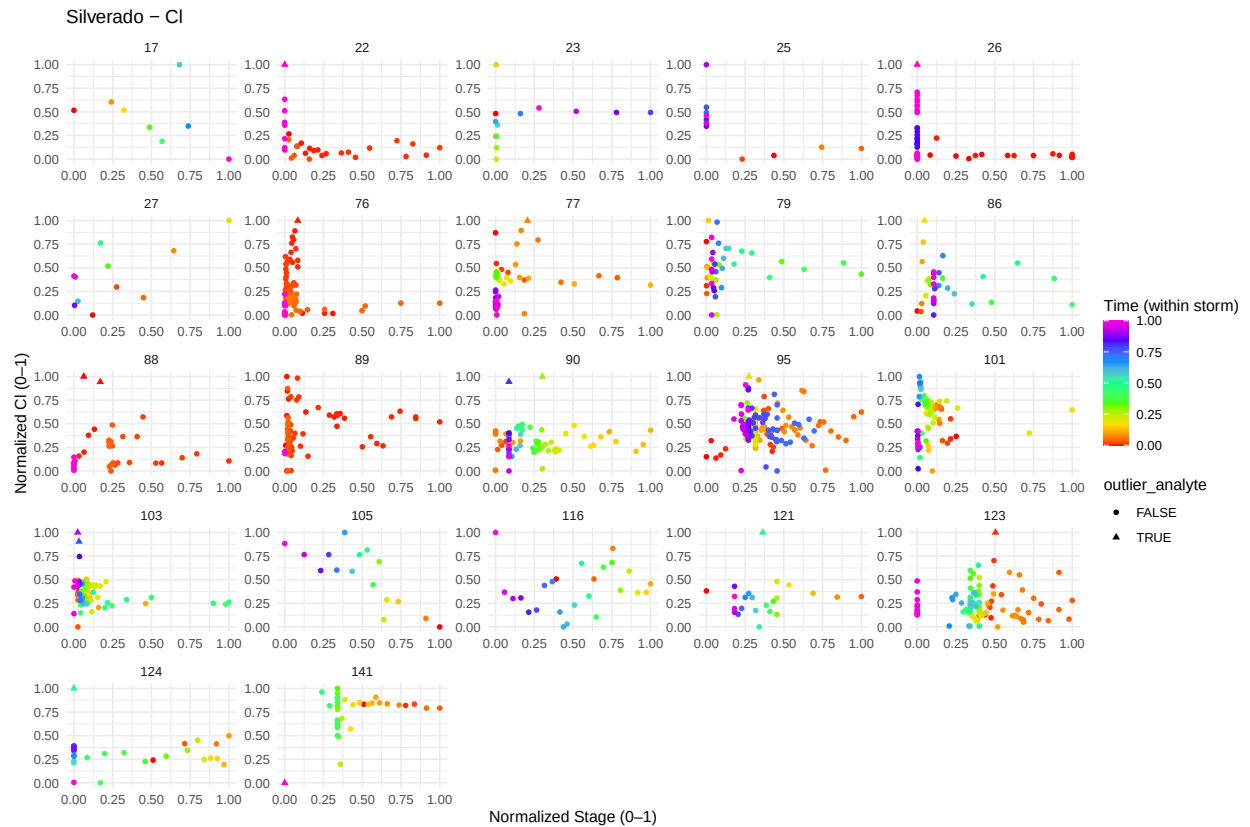
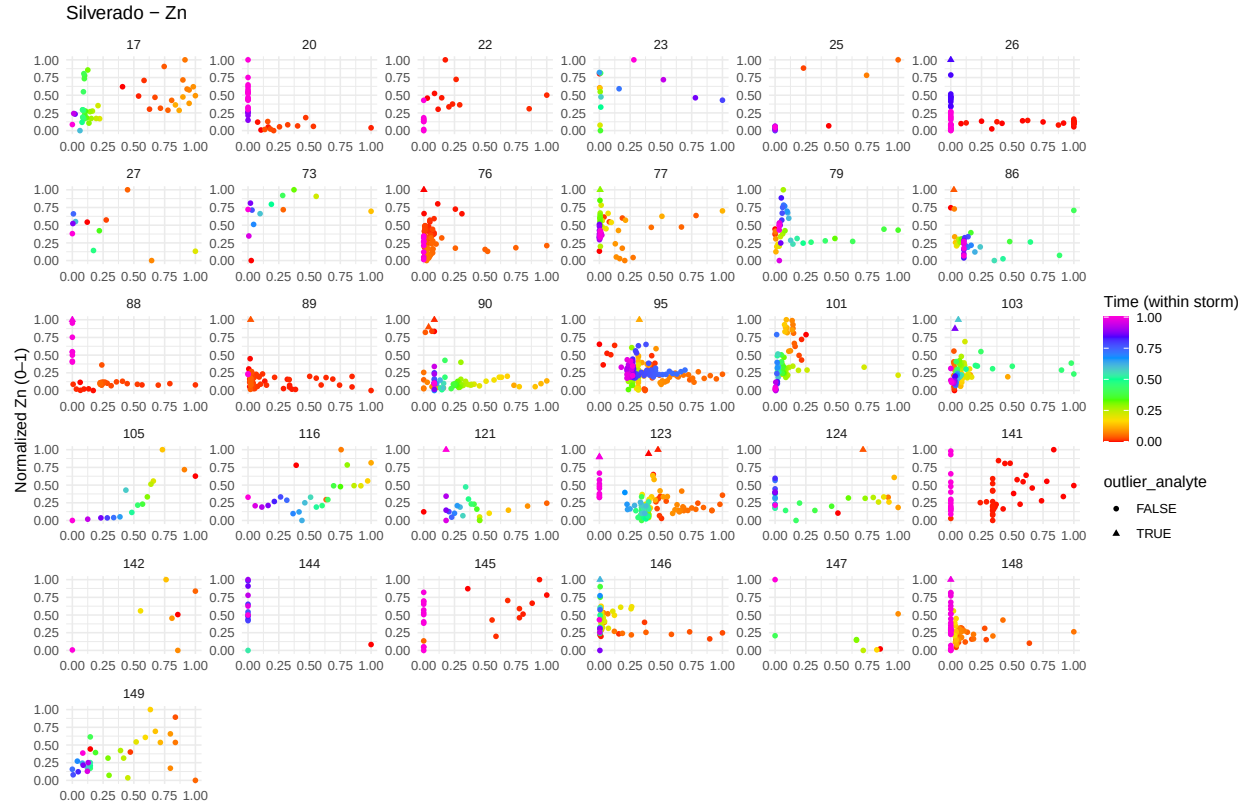


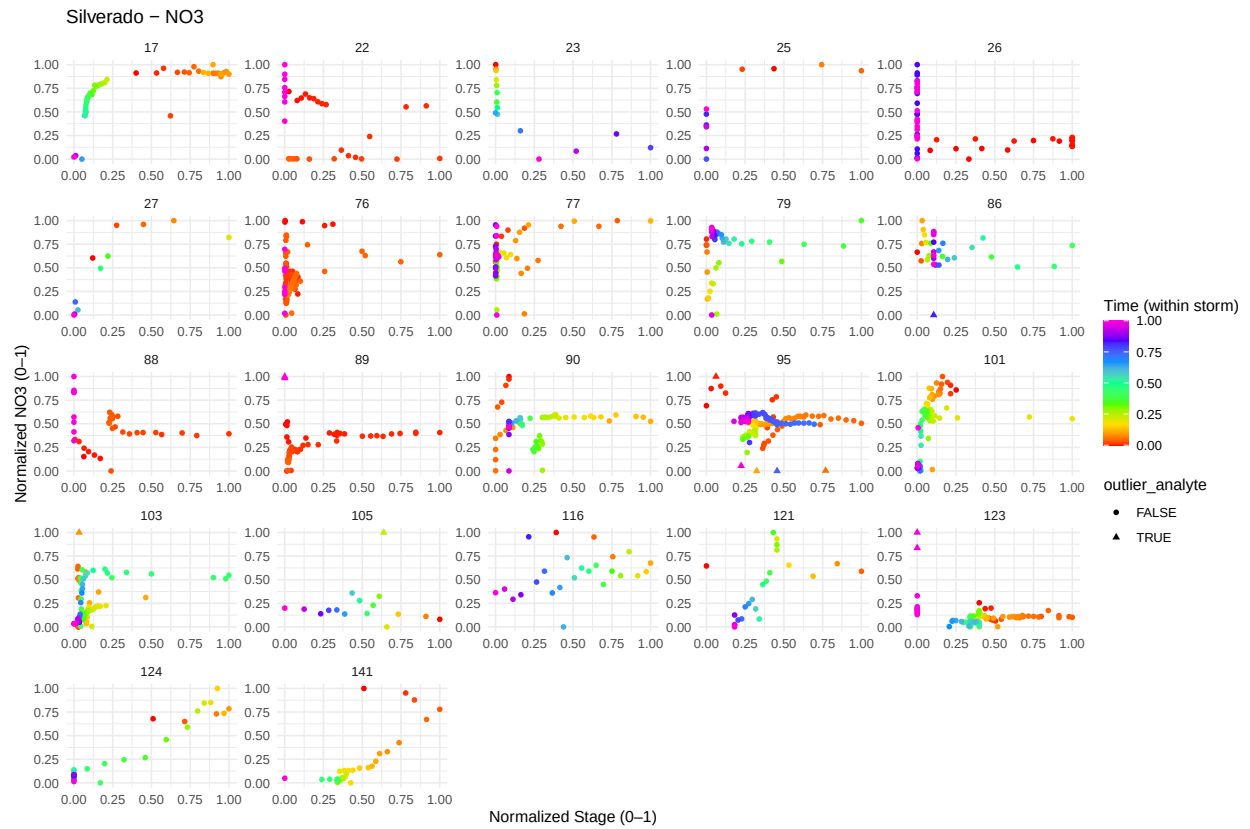
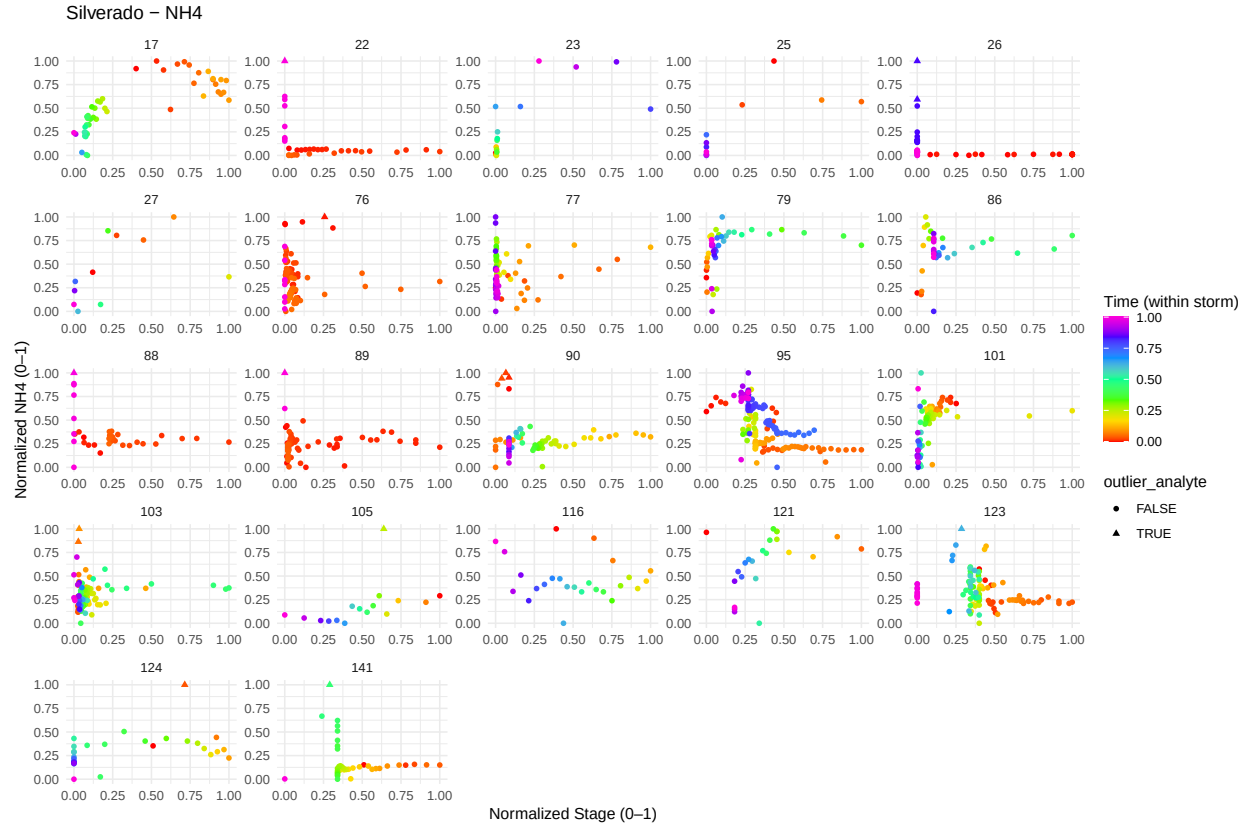


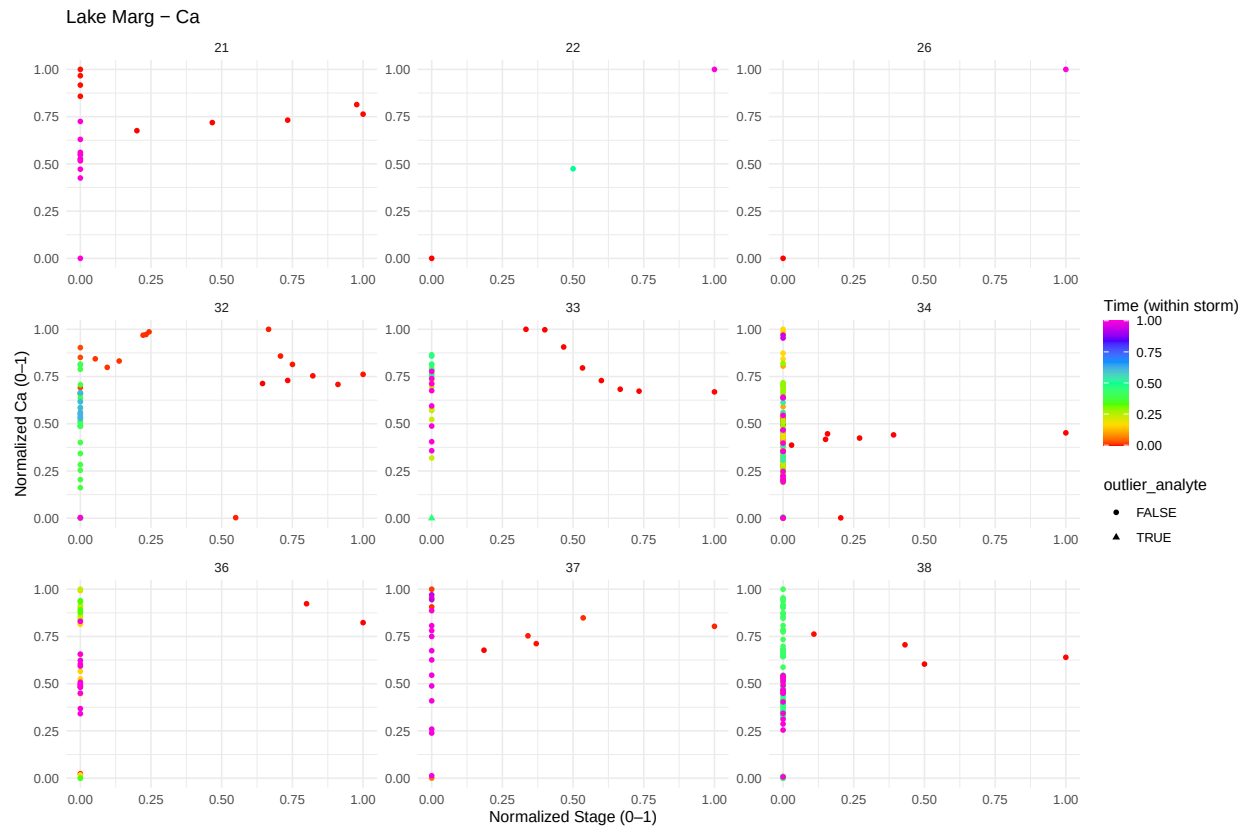
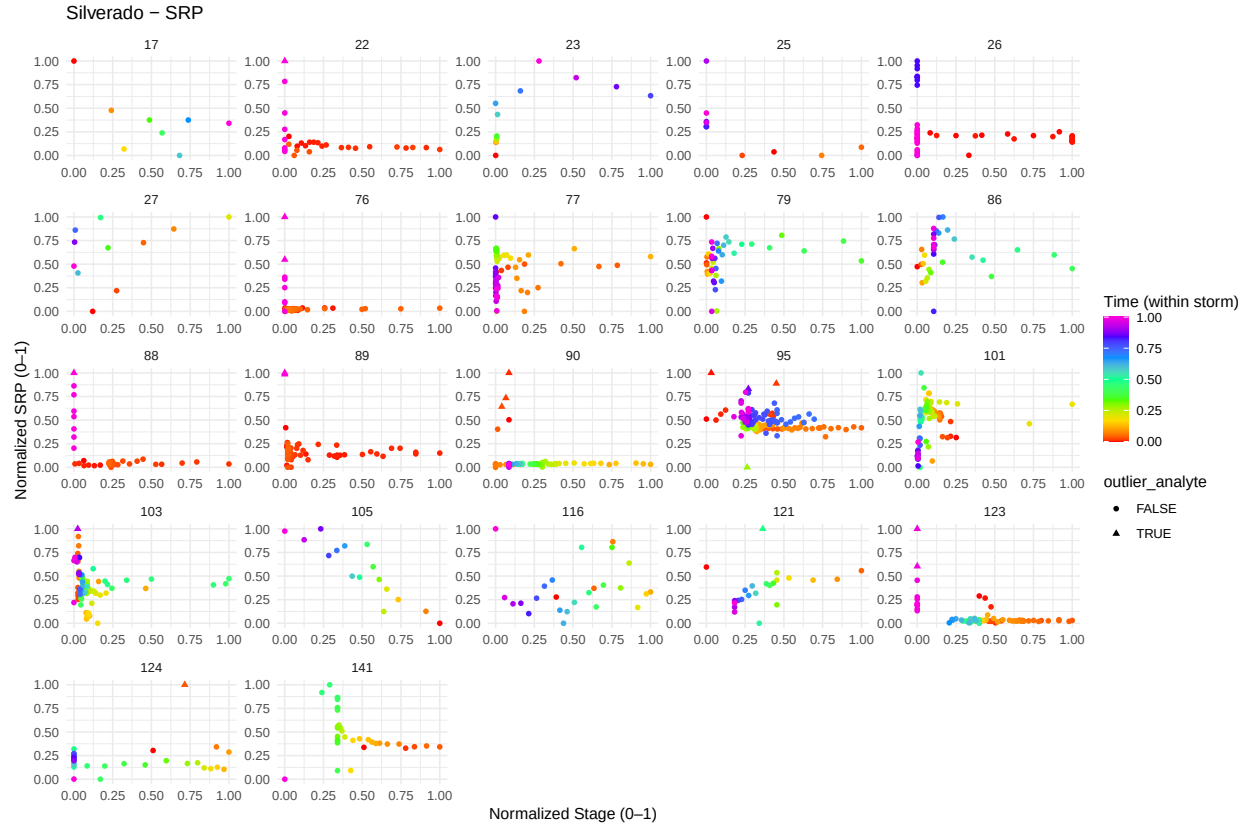


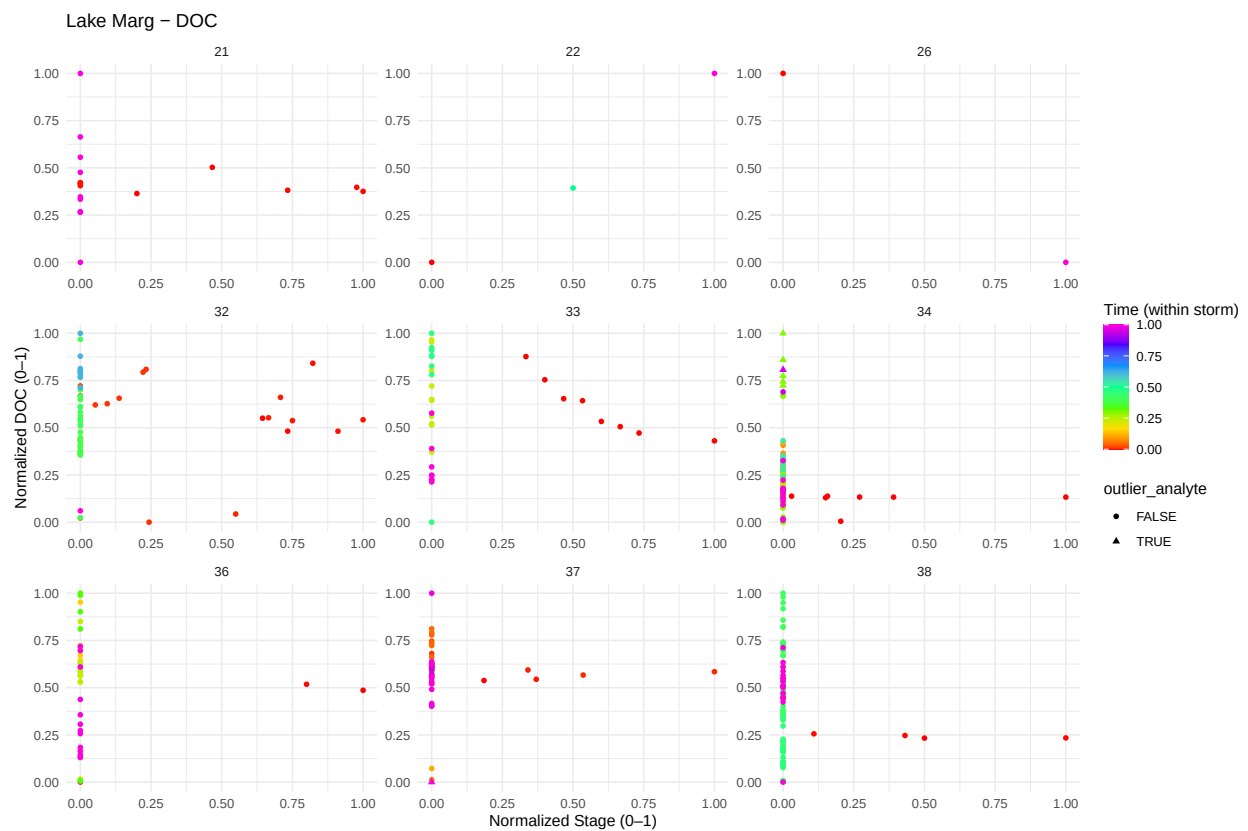
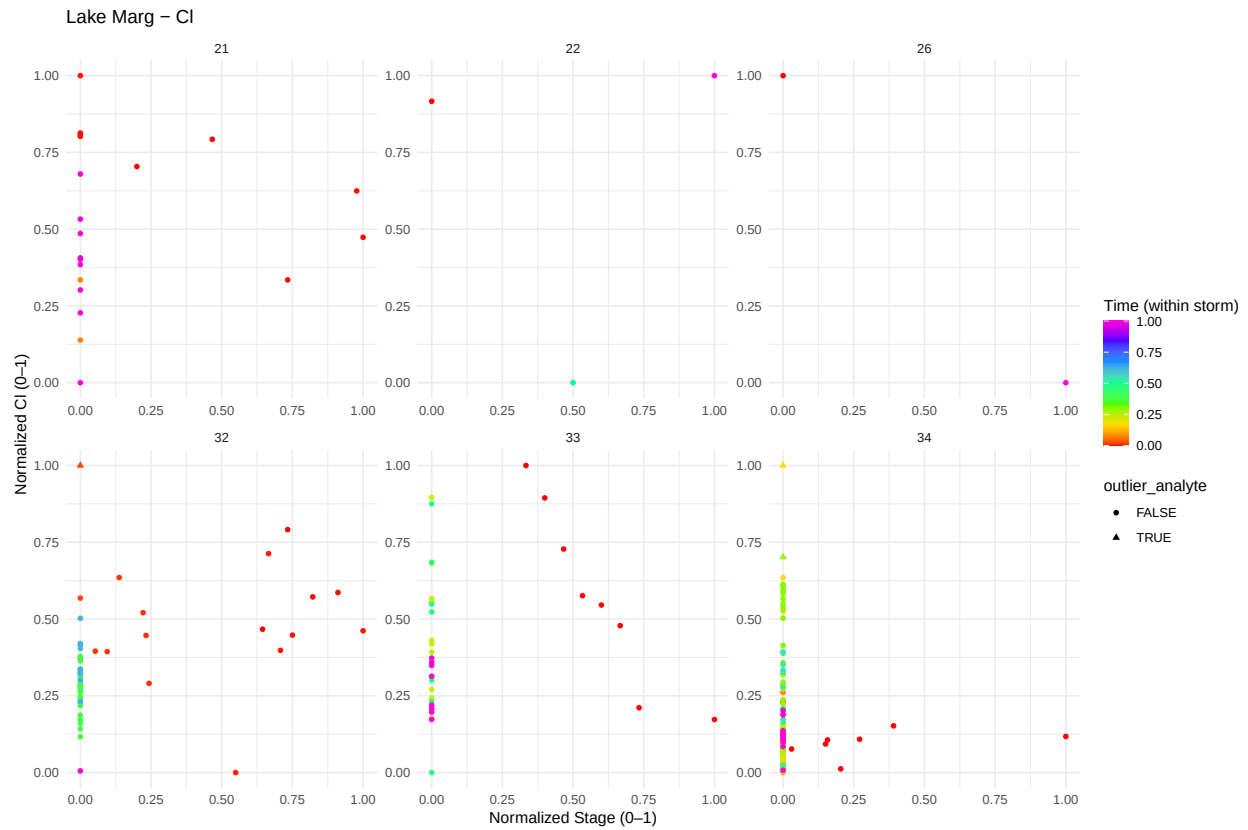


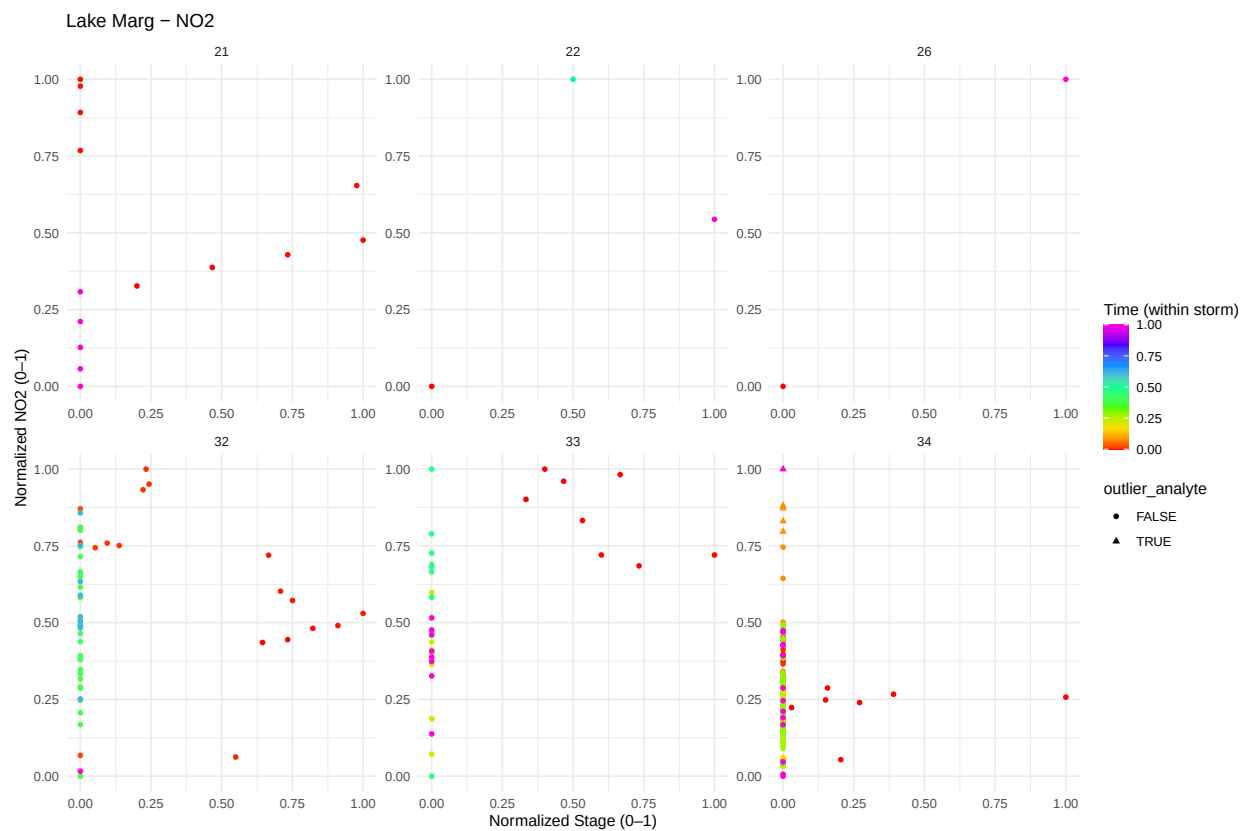
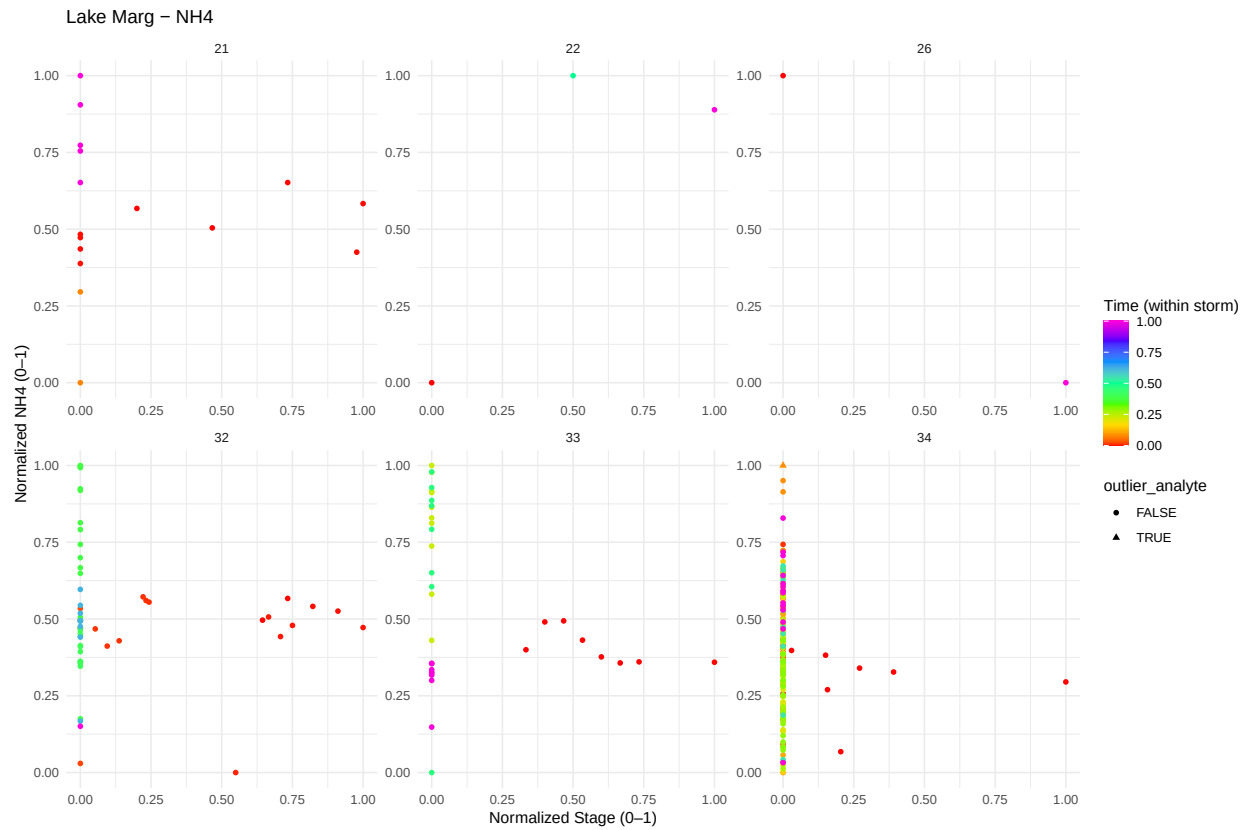


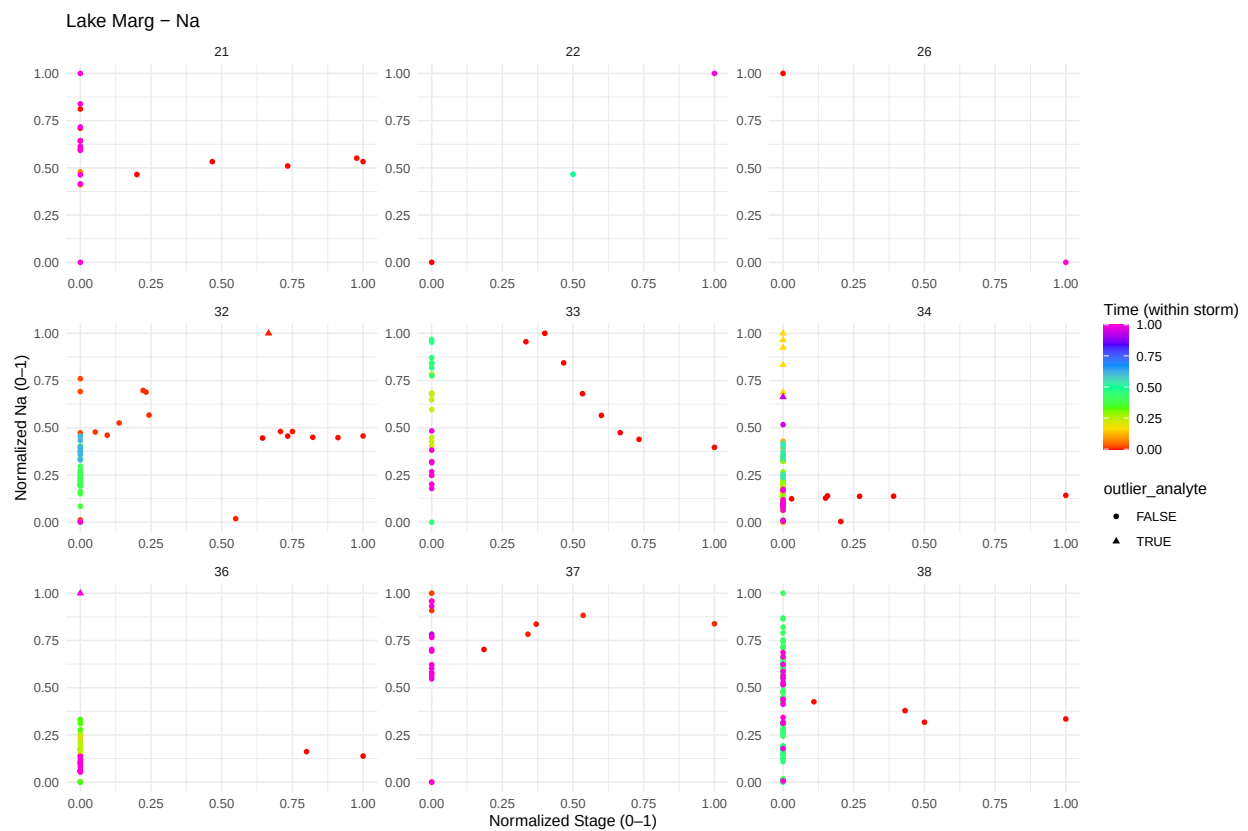
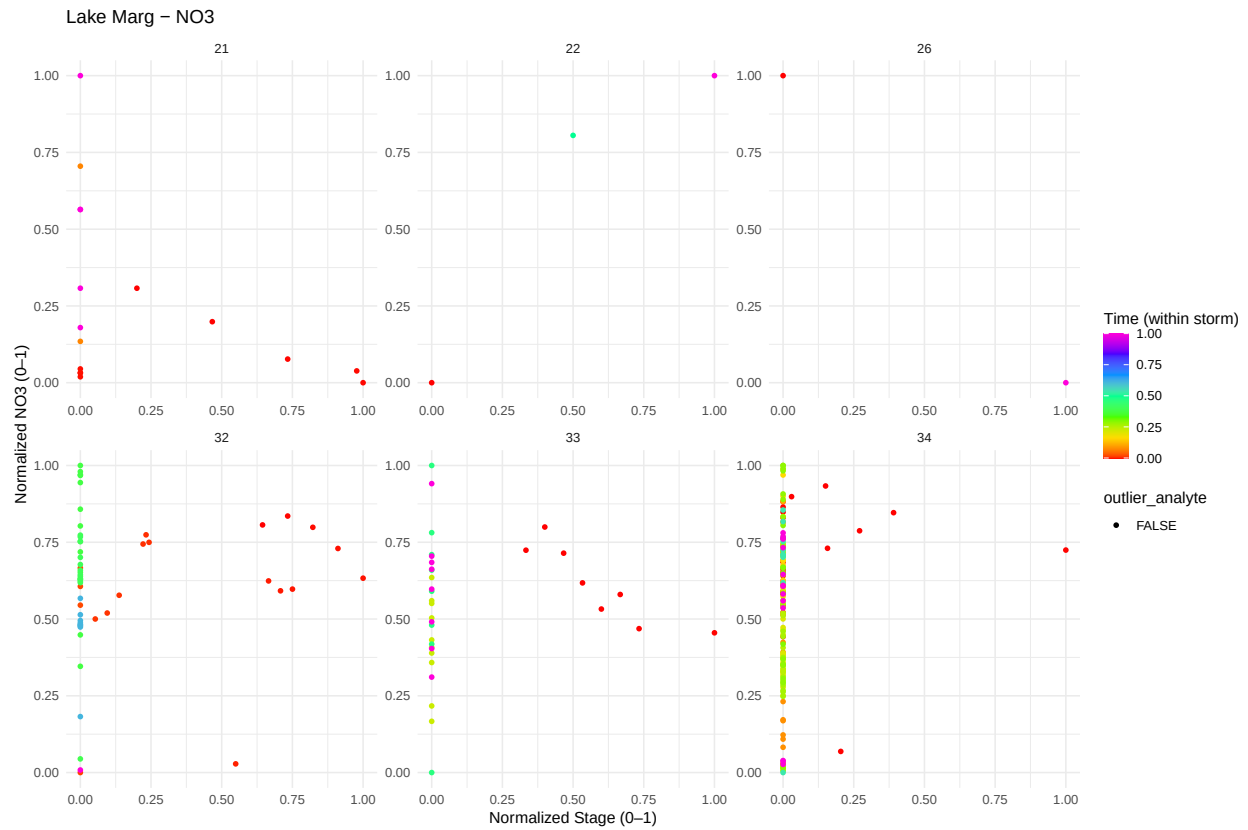


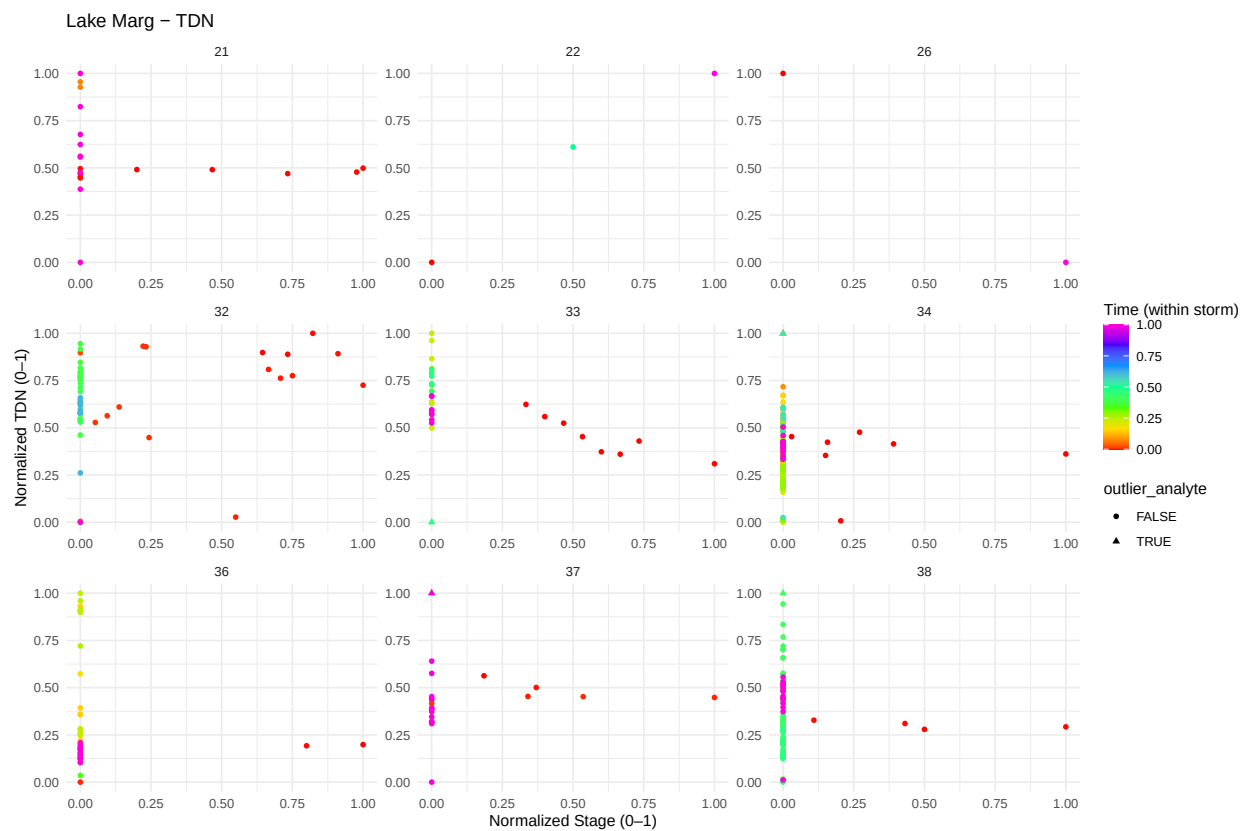
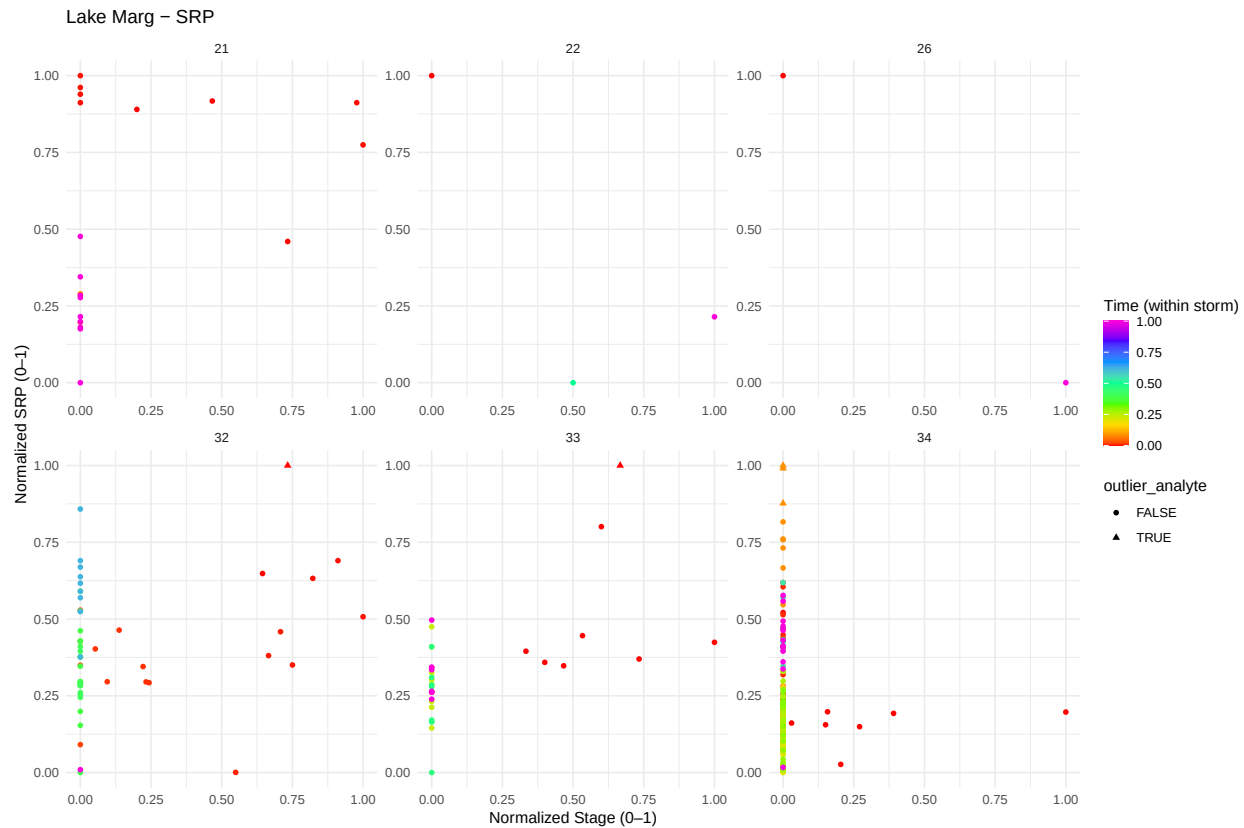




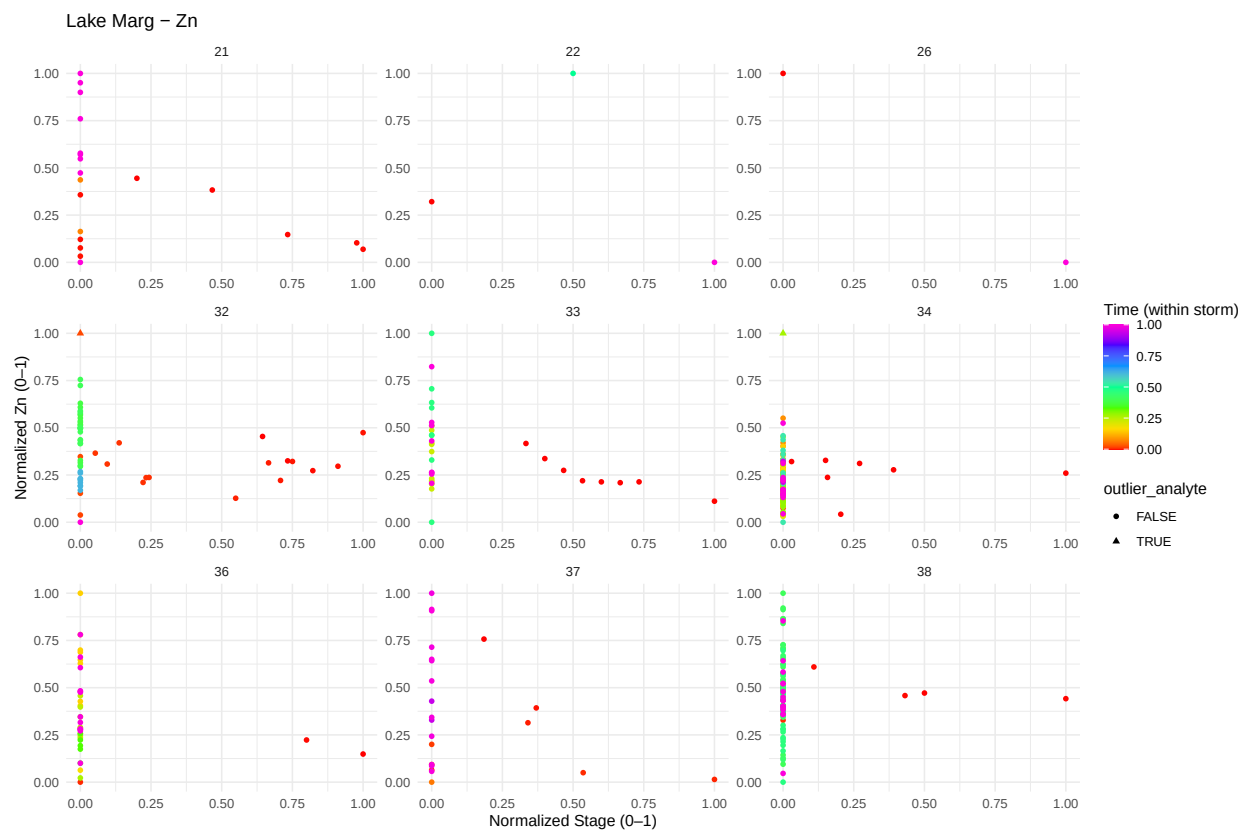
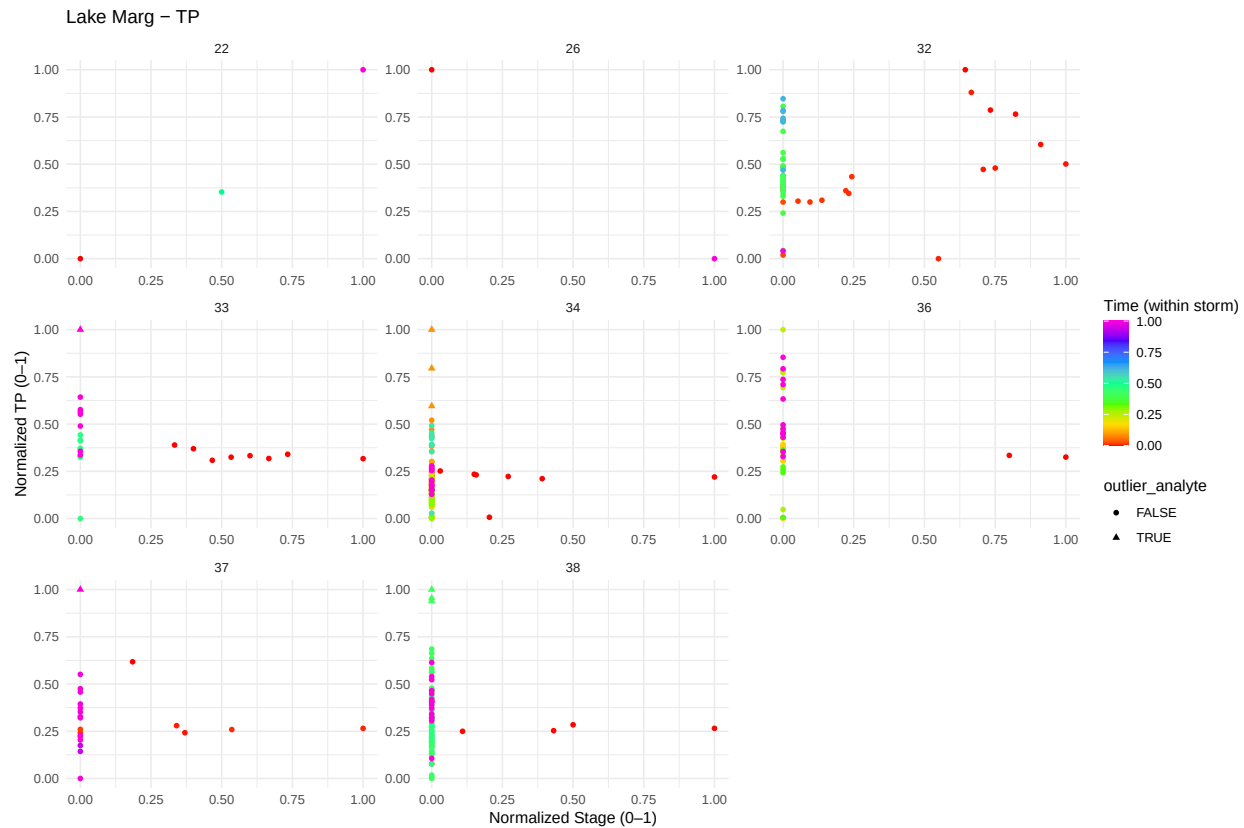












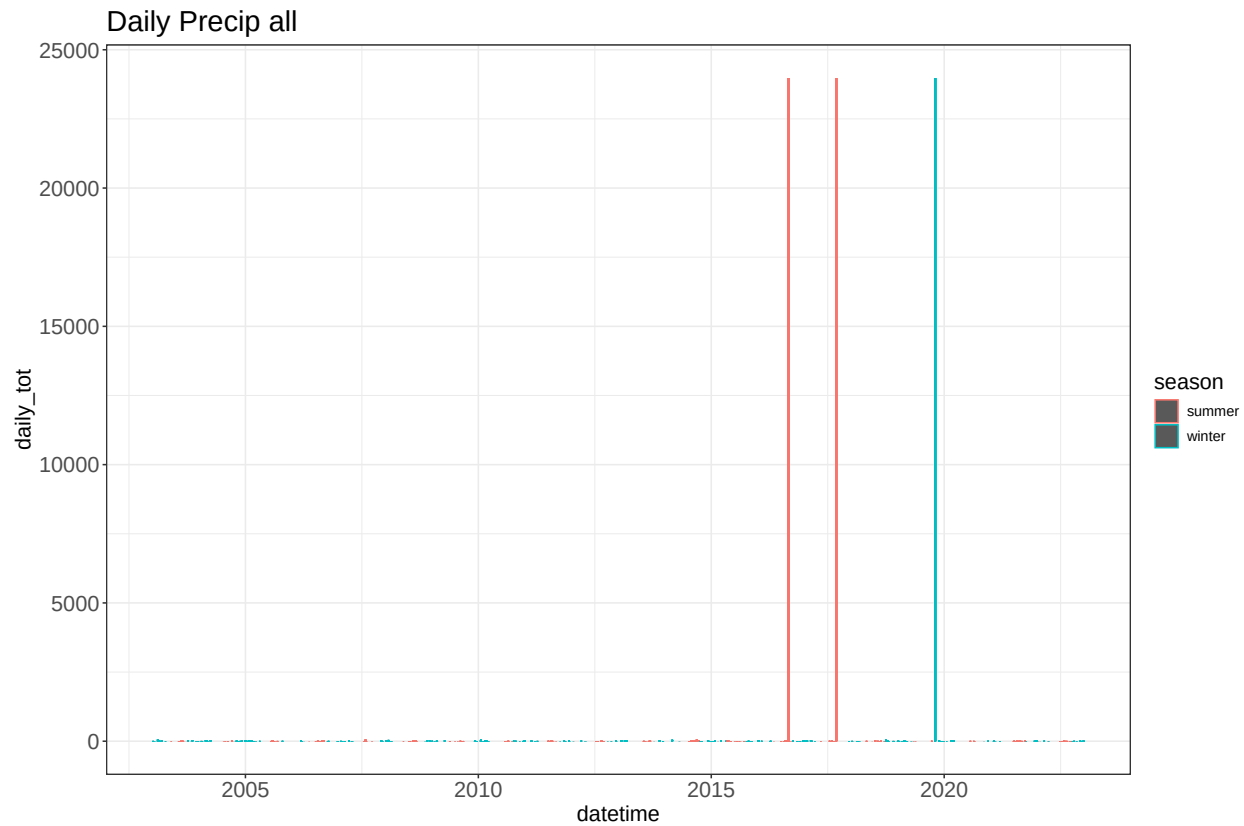
## Precipitation

Precipitation data was collected from <https://azmet.arizona.edu/azmet/27.htm> - precipitation gage at the Desert Ridge station

Time series of precip

Daily Precip (excluding large outliers)

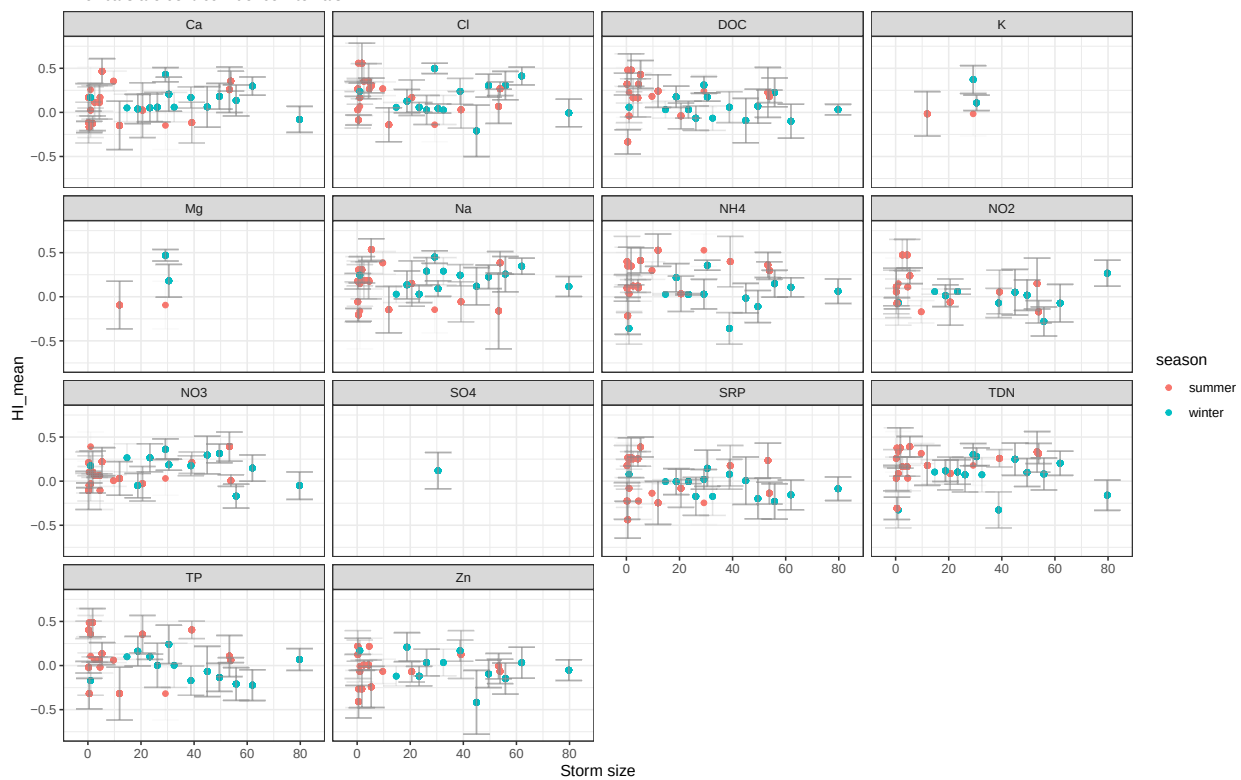




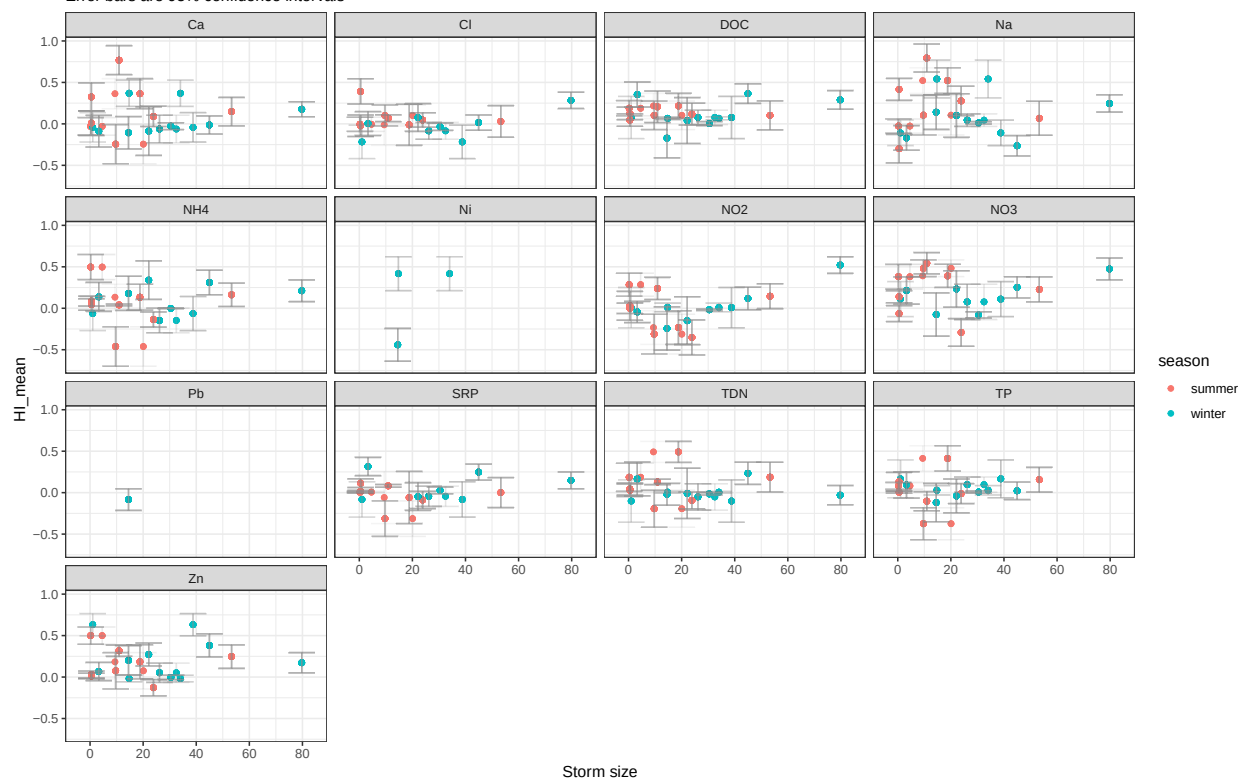
Storm size = cumulative precipitation (mm) per storm

Curry HI mean vs cumulative precip per storm

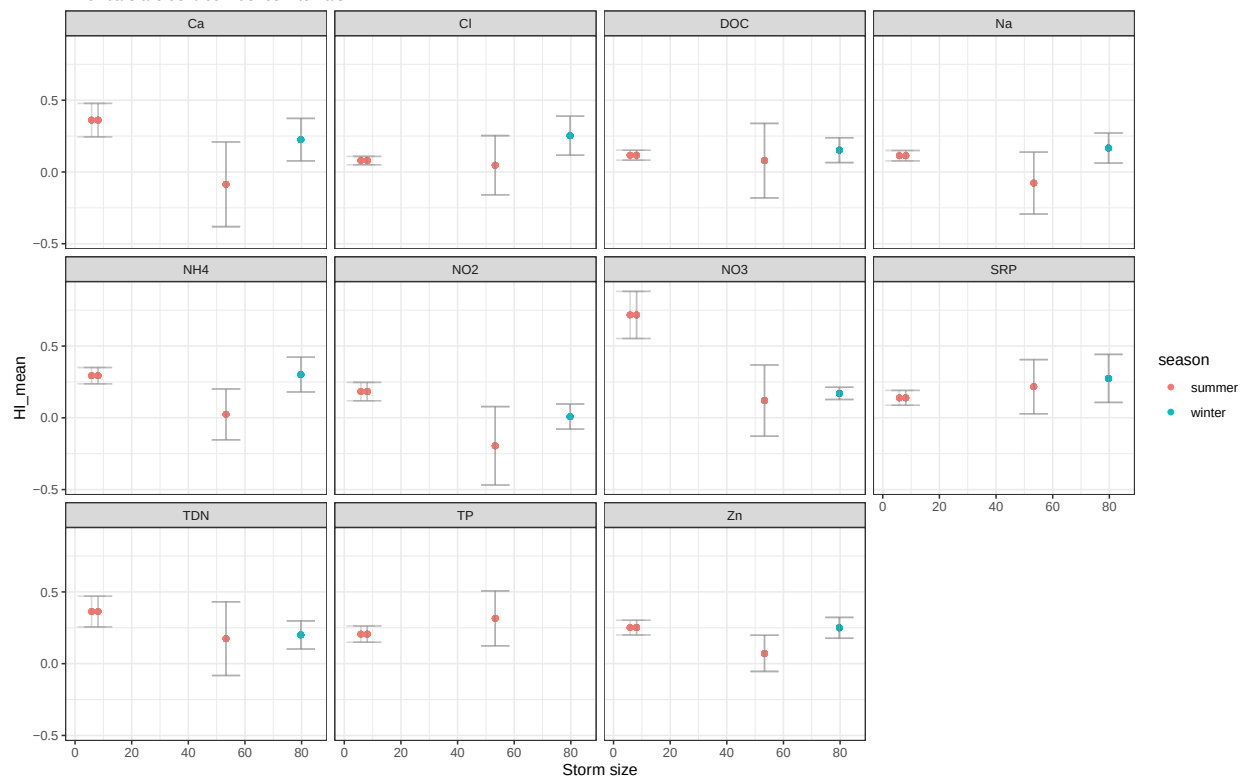
Error bars are 95% confidence intervals



Silverado HI mean vs cumulative precip per storm  
Error bars are 95% confidence intervals

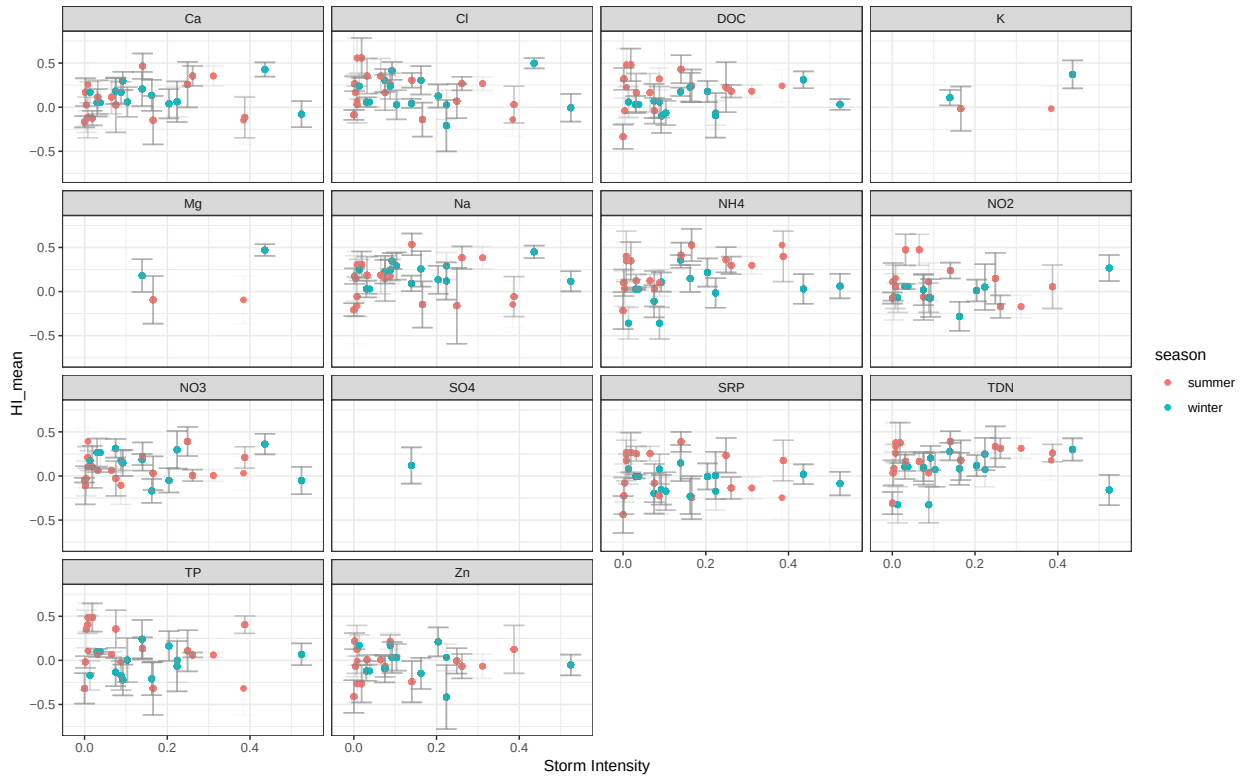


Lake Margarita HI mean vs cumulative precip per storm  
Error bars are 95% confidence intervals

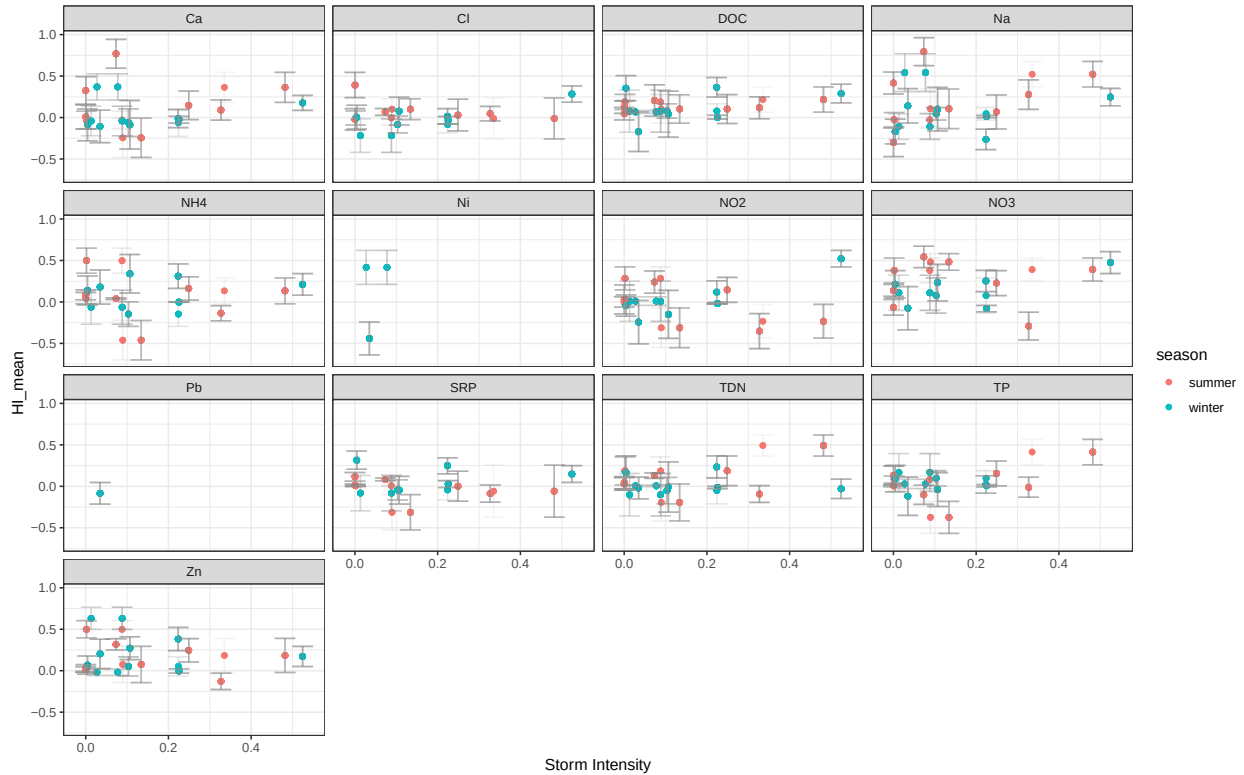


Storm intensity = cumulative precipitation/storm duration

Curry HI mean vs storm intensity  
Error bars are 95% confidence intervals



Silverado HI mean vs storm intensity  
Error bars are 95% confidence intervals



# Lake Margarita HI mean vs storm intensity

Error bars are 95% confidence intervals

